

Lesson 03 Demo 06

Adding Microsoft Entra ID Authentication for Storage

Objective: To create a guest user in Microsoft Entra ID and provide relevant access to the user, allowing them to access Azure storage accounts using Storage Explorer

Tools required: Azure account with administrator access

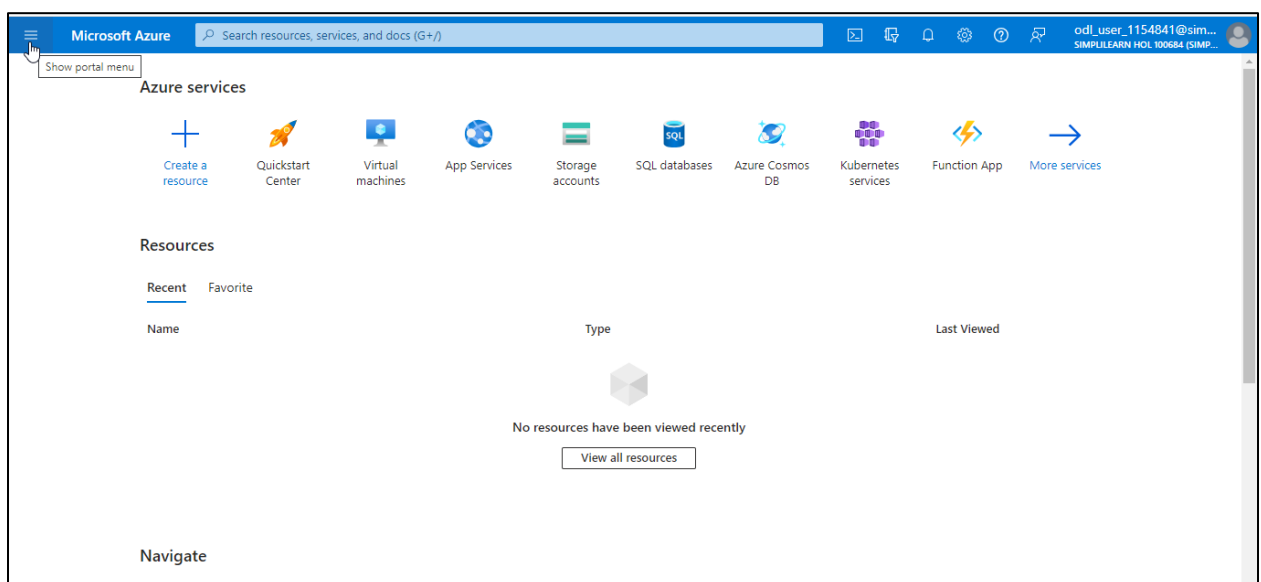
Prerequisites: None

Steps to be followed:

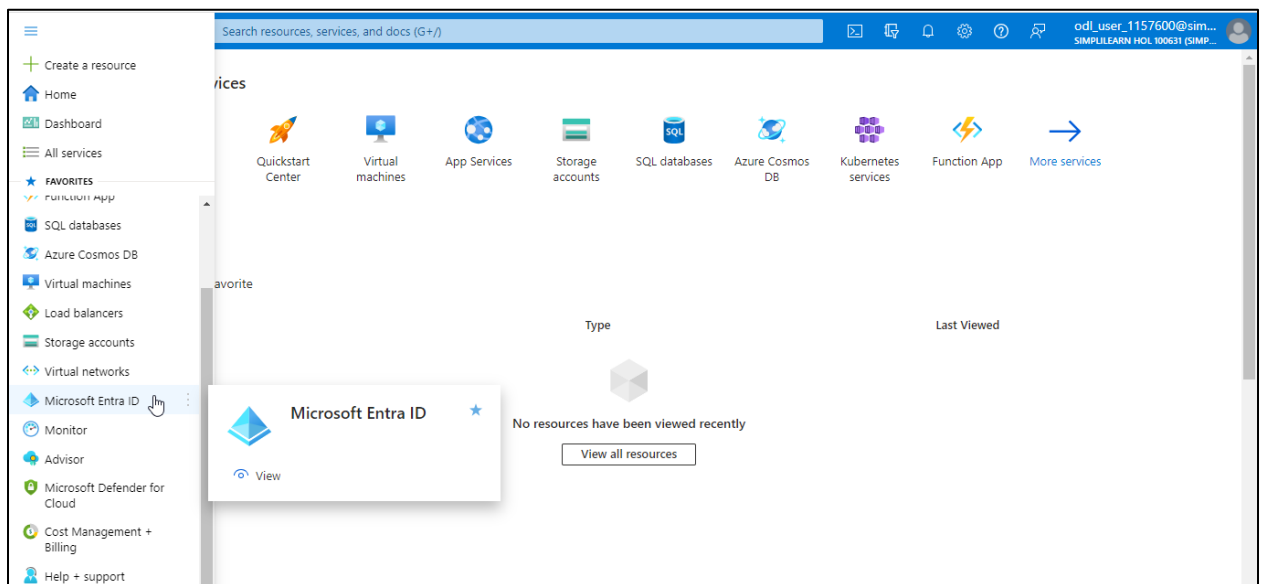
1. Create a new user in Microsoft Entra ID
2. Create an Azure storage account and upload a blob
3. Download Azure Storage Explorer
4. Add Microsoft Entra ID authentication for storage

Step 1: Create a new user in Microsoft Entra ID

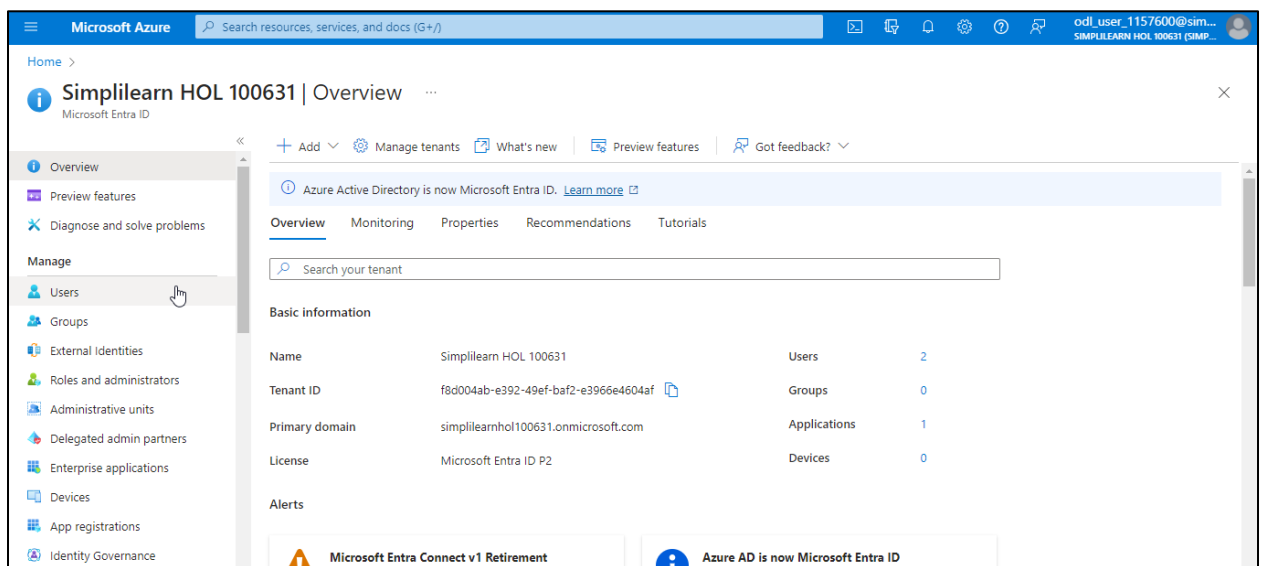
- 1.1 Navigate to the **Microsoft Azure** home page and click on the three lines located on the left side of the page



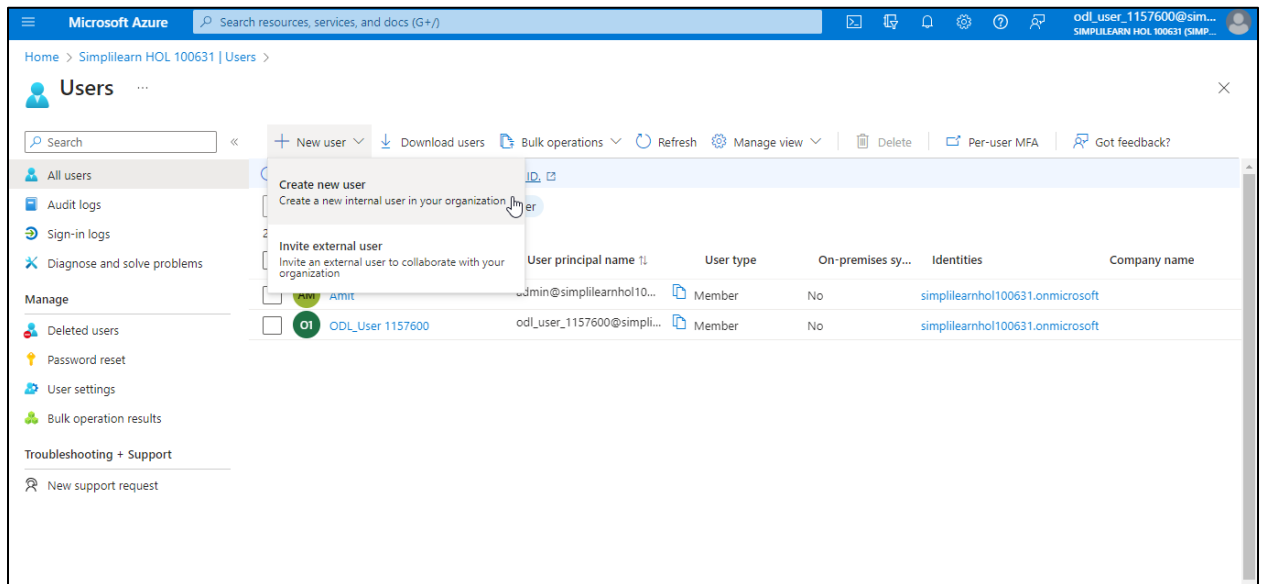
1.2 Select Microsoft Entra ID



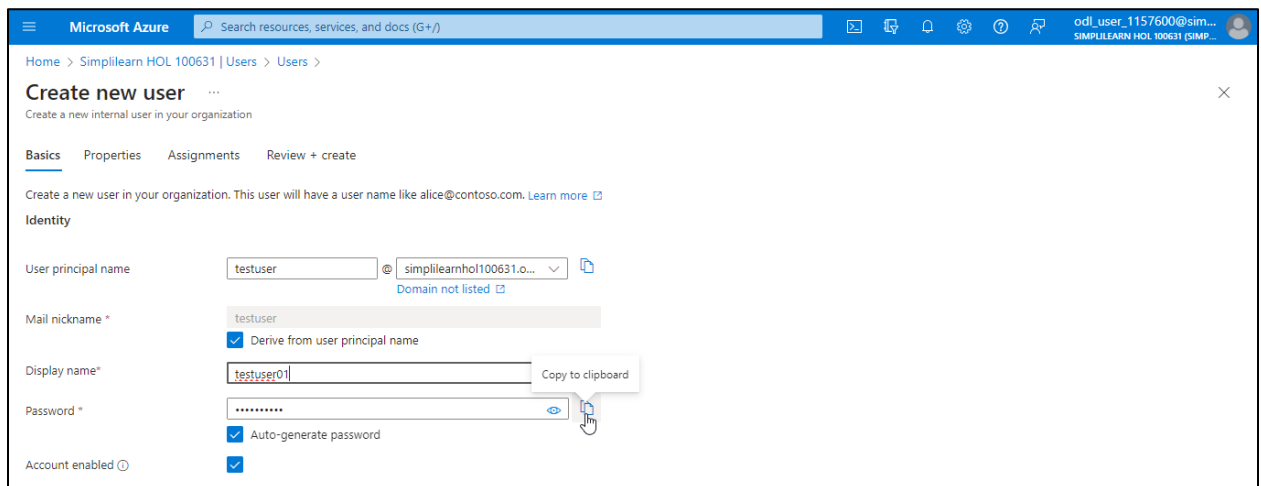
1.3 In the Overview page, click on Users under the Manage section



1.4 In the **Users** page, click on **+ New user** and select **Create new user**



1.5 In the **Create new user** page, navigate to the **Basics** tab, and provide the details as shown below:



Note: Copy the auto-generated password, as you need to log in with the same credentials to test the Microsoft Entra ID Storage authentication.

1.6 Click Review + create

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Simplilearn HOL 100631 | Users > Users >

Create new user

Create a new internal user in your organization

Basics Properties Assignments **Review + create**

Create a new user in your organization. This user will have a user name like alice@contoso.com. [Learn more](#)

Identity

User principal name: testuser @ simplilearnhol100631.o... [Domain not listed](#)

Mail nickname *: testuser ☒ Derive from user principal name

Display name*: testuser01

Password *: ☒ Auto-generate password

Account enabled ☒

Review + create < Previous Next: Properties > [Give feedback](#)

1.7 Click Create

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Simplilearn HOL 100631 | Users > Users >

Create new user

Create a new internal user in your organization

Basics Properties Assignments **Review + create**

Basics

User principal name: testuser@simplilearnhol100631.onmicrosoft.com

Display name: testuser01

Mail nickname: testuser

Password: ☐

Account enabled: Yes

Properties

User type: Member

Assignments

Administrative units

Create < Previous Next > [Give feedback](#)

The test user has been successfully created as shown below:

Microsoft Azure portal interface showing the 'Users' page. A notification banner at the top right indicates 'Successfully created user' and 'Successfully created user testuser01'. The table displays 2 users found:

Display name	User principal name	User type	On-premises sy...	Identities	Company name
Amit	admin@simpilearnhol10...	Member	No	simpilearnhol100631.onmicrosoft	
ODL_User 1157600	odl_user_1157600@simpli...	Member	No	simpilearnhol100631.onmicrosoft	

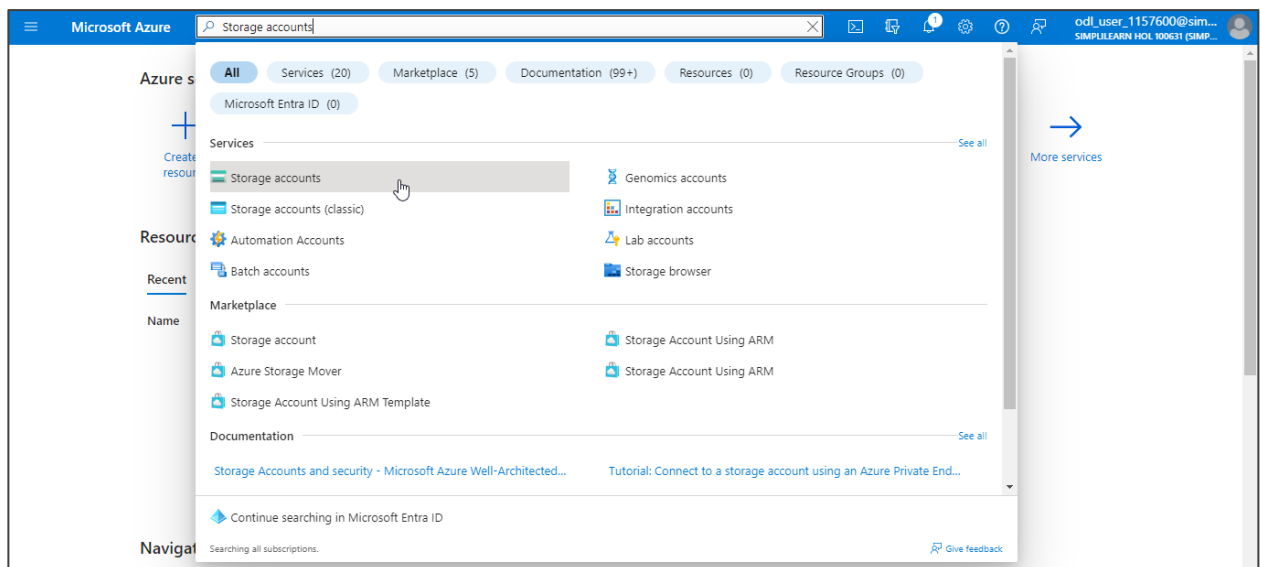
1.8 Click on the **Refresh** button to view the created test user (testuser01)

Microsoft Azure portal interface showing the 'Users' page after clicking the 'Refresh' button. The table now displays 3 users found, including the newly created 'testuser01':

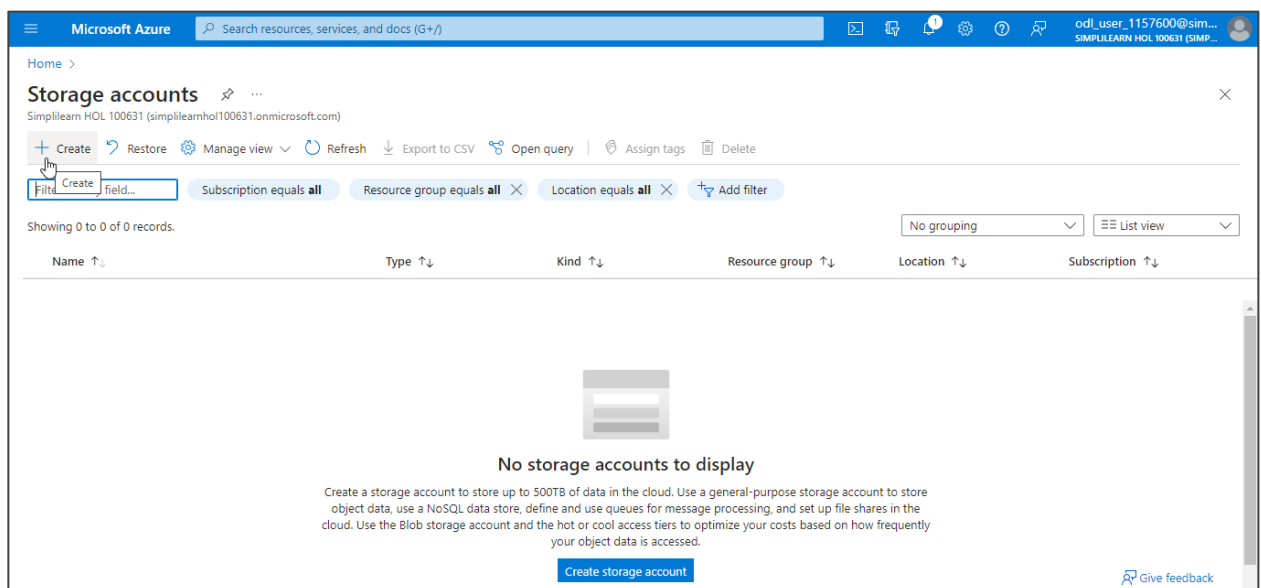
Display name	User principal name	User type	On-premises sy...	Identities	Company name
Amit	admin@simpilearnhol10...	Member	No	simpilearnhol100631.onmicrosoft	
ODL_User 1157600	odl_user_1157600@simpli...	Member	No	simpilearnhol100631.onmicrosoft	
testuser01	testuser@simpilearnhol1...	Member	No	simpilearnhol100631.onmicrosoft	

Step 2: Create an Azure storage account and upload a blob

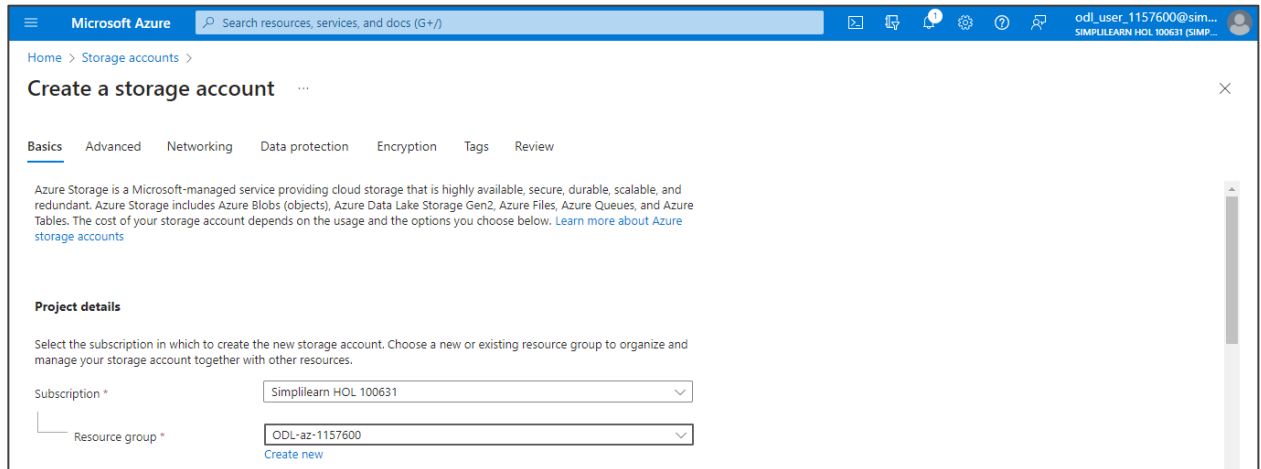
2.1 In the Microsoft Azure home page, search for and select **Storage accounts**



2.2 In the **Storage accounts** page, click on **+ Create**



2.3 In the **Create a storage account** page, navigate to the **Basics** tab and provide the following information under **Project details** as shown below:

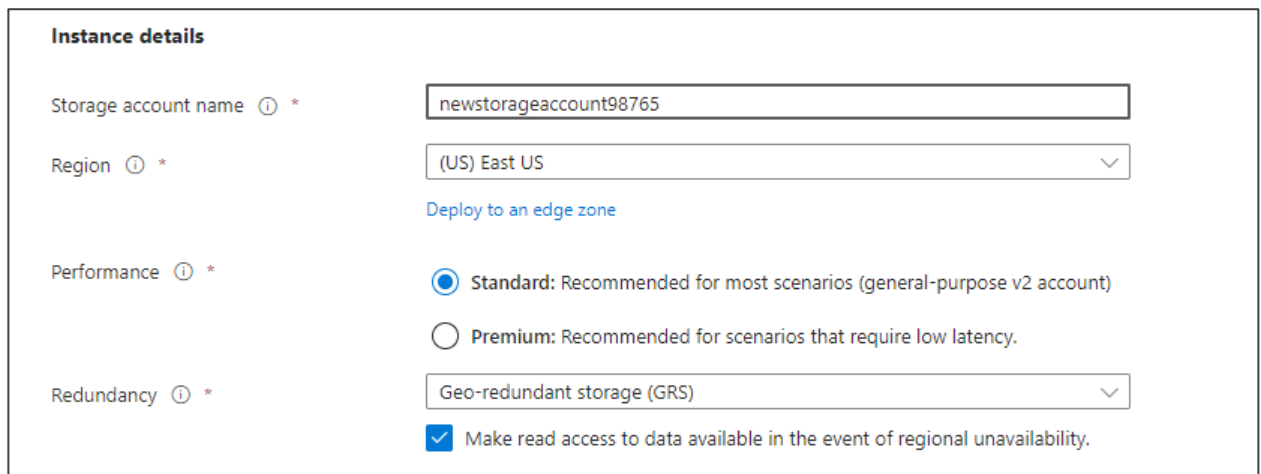


The screenshot shows the 'Create a storage account' page in the Microsoft Azure portal. The 'Basics' tab is selected. Under the 'Project details' section, the following information is provided:

- Subscription ***: Simplilearn HOL 100631
- Resource group ***: ODL-az-1157600

Below the resource group dropdown, there is a link that says 'Create new'.

2.4 Under **Instance details**, provide the information as shown below:



The screenshot shows the 'Instance details' section of the 'Create a storage account' page. The following information is provided:

- Storage account name ***: newstorageaccount98765
- Region ***: (US) East US
- Performance ***: ☒ **Standard**: Recommended for most scenarios (general-purpose v2 account). ☐ **Premium**: Recommended for scenarios that require low latency.
- Redundancy ***: Geo-redundant storage (GRS)
- ☒ **Make read access to data available in the event of regional unavailability.**

2.5 Click Review

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Storage accounts >

Create a storage account

Basics Advanced Networking Data protection Encryption Tags **Review**

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below. [Learn more about Azure storage accounts](#)

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *

Resource group * [Create new](#)

Review < Previous Next : Advanced >

[Give feedback](#)

2.6 Click Create

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Storage accounts >

Create a storage account

Basics Advanced Networking Data protection Encryption Tags **Review**

Basics

Subscription	Simplilearn HOL 100631
Resource Group	ODL-az-1157600
Location	eastus
Storage account name	newstorageaccount98765
Deployment model	Resource manager
Performance	Standard
Replication	Read-access geo-redundant storage (RA-GRS)

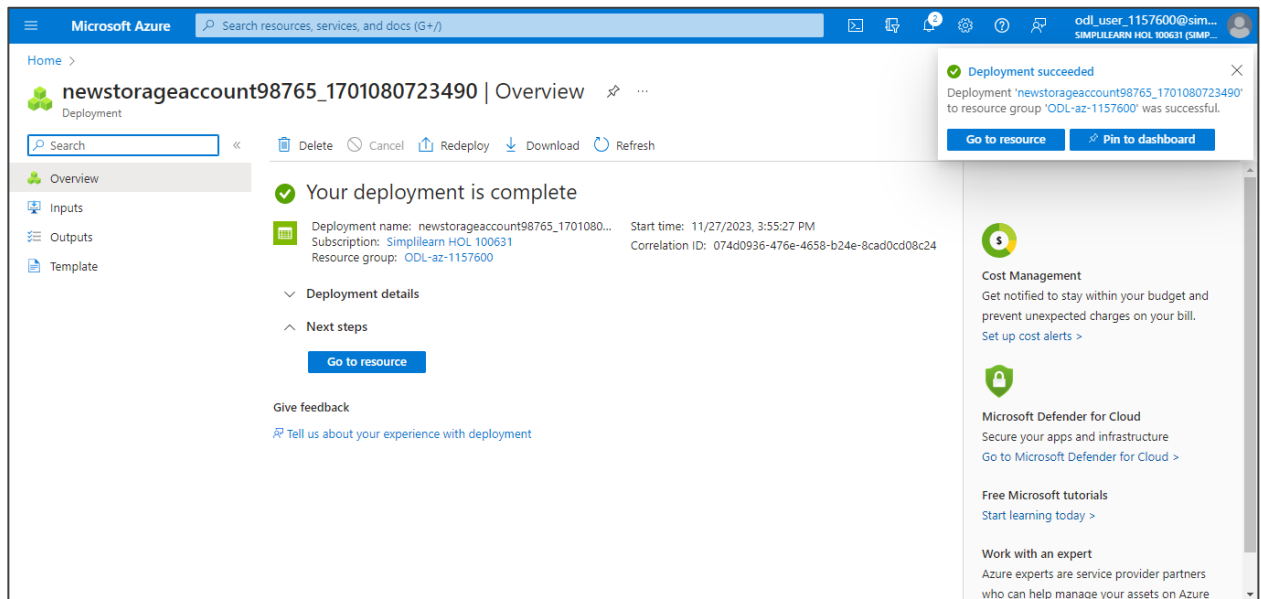
Advanced

Enable hierarchical namespace	Disabled
Enable network file system v3	Disabled

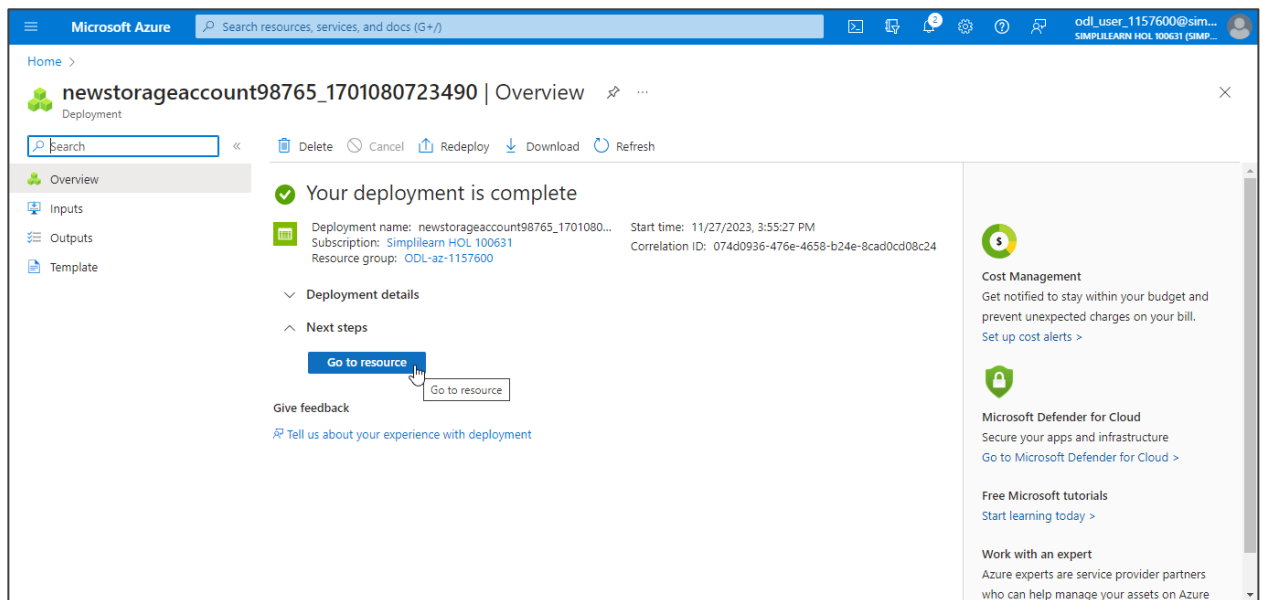
Create < Previous Next > [Download a template for automation](#)

[Give feedback](#)

The deployment has been completed as shown below:



2.7 Click Go to resource



2.8 Click on **Containers** under **Data storage**

newstorageaccount98765
Storage account

Search resources, services, and docs (G+/I)

Upload Open in Explorer Delete Move Refresh Open in mobile CLI / PS Feedback

Essentials JSON View

Resource group (move) : ODL-az-1157600
Location : eastus
Primary/Secondary Location : Primary: East US, Secondary: West US
Subscription (move) : Simplilearn HOL 100631
Subscription ID : 0a2e60a1-8123-4b36-b88a-3e4b4ae18379
Disk state : Primary: Available, Secondary: Available

Performance : Standard
Replication : Read-access geo-redundant storage (RA-GRS)
Account kind : StorageV2 (general purpose v2)
Provisioning state : Succeeded
Created : 11/27/2023, 3:55:34 PM

Tags (edit) : Add tags

Properties Monitoring Capabilities (7) Recommendations (0) Tutorials Tools + SDKs

Blob service

Hierarchical namespace : Disabled
Default access tier : Hot
Blob anonymous access : Disabled
Blob soft delete : Enabled (7 days)

Security

Require secure transfer for REST API operations : Enabled
Storage account key access : Enabled
Minimum TLS version : Version 1.2

2.9 On the **Containers** page, click **+ Container**

newstorageaccount98765 | Containers

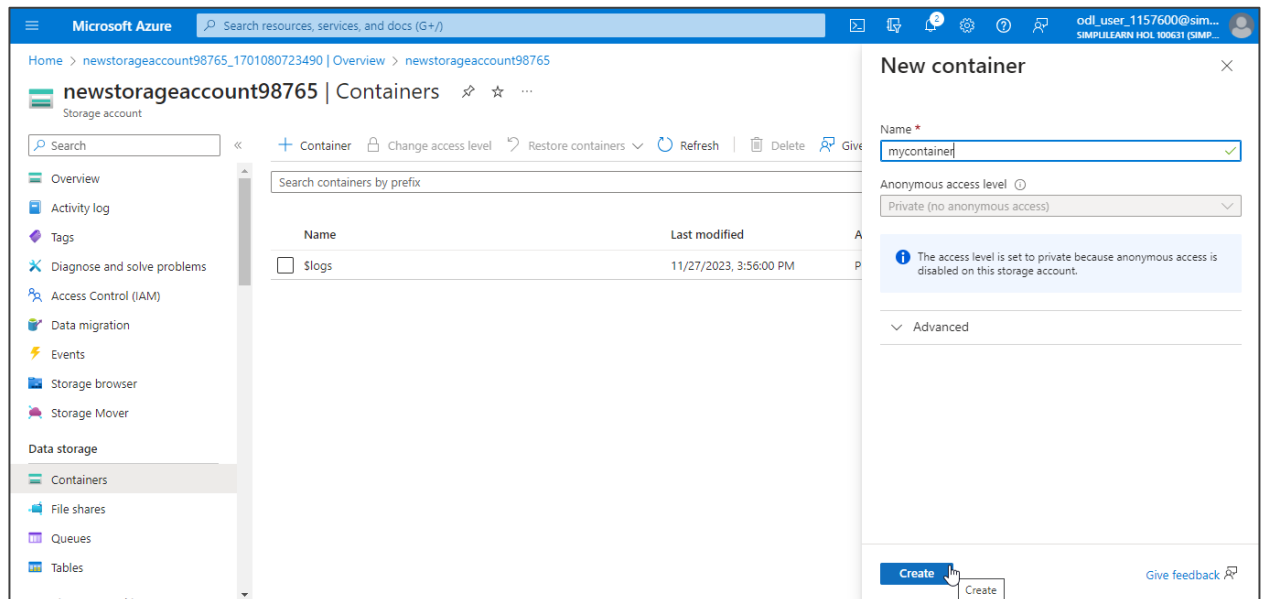
Search resources, services, and docs (G+/I)

+ Container Change access level Restore containers Refresh Delete Give feedback

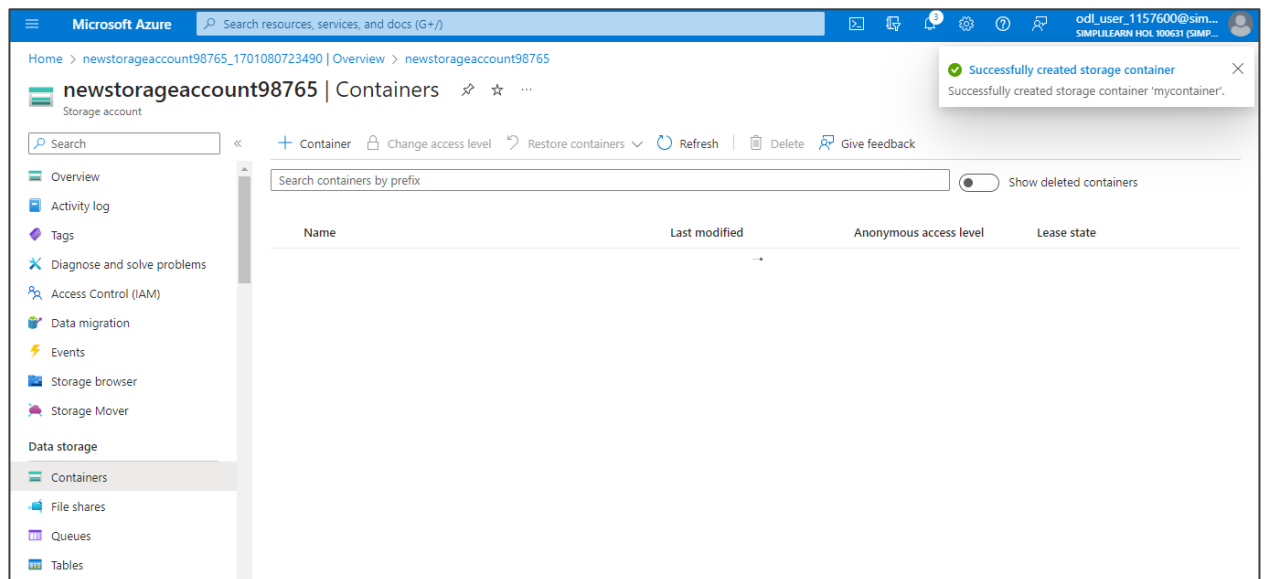
Search cd New container [x] Show deleted containers

Name	Last modified	Anonymous access level	Lease state
☐ \$logs	11/27/2023, 3:56:00 PM	Private	Available

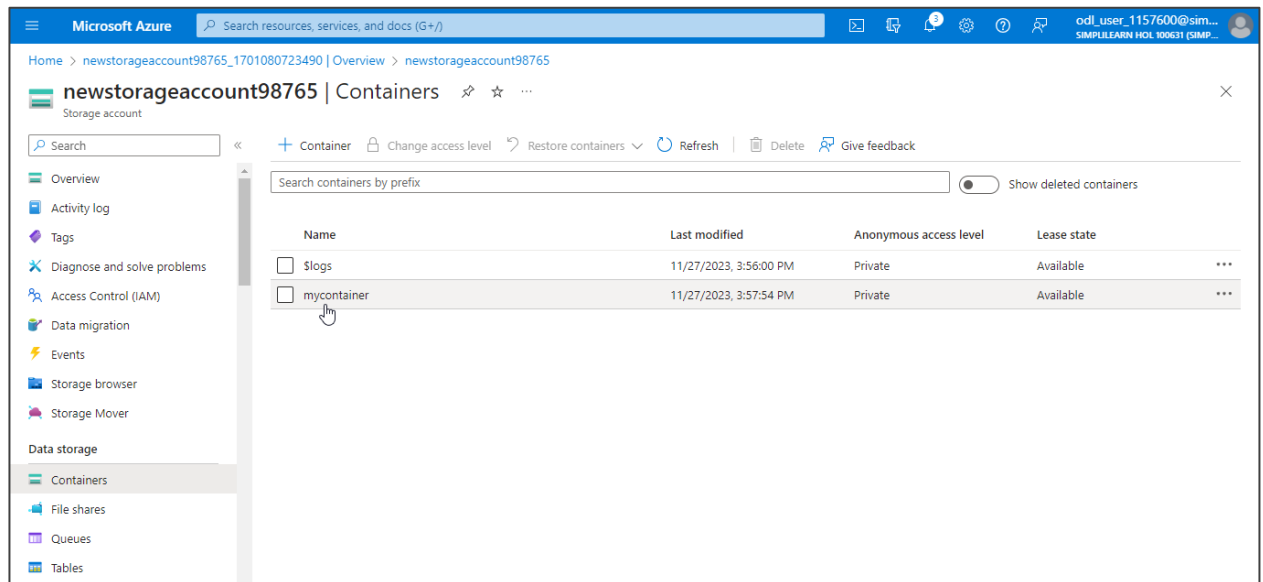
2.10 In the **New container** page, provide a **Name (mycontainer)** and click **Create**



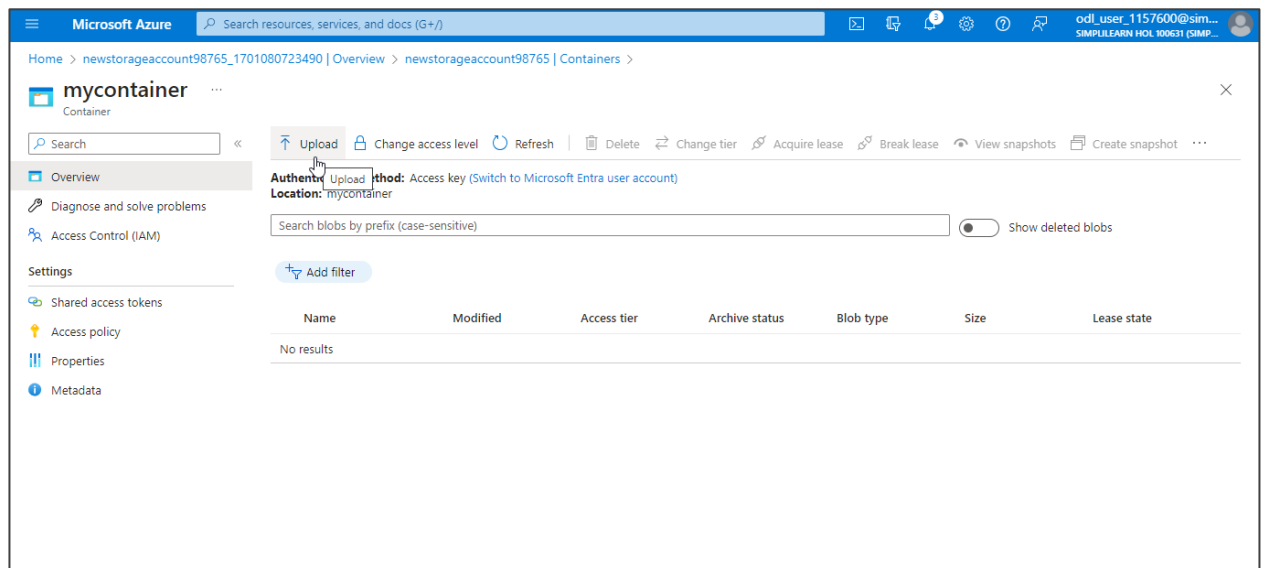
The container has been created as shown below:



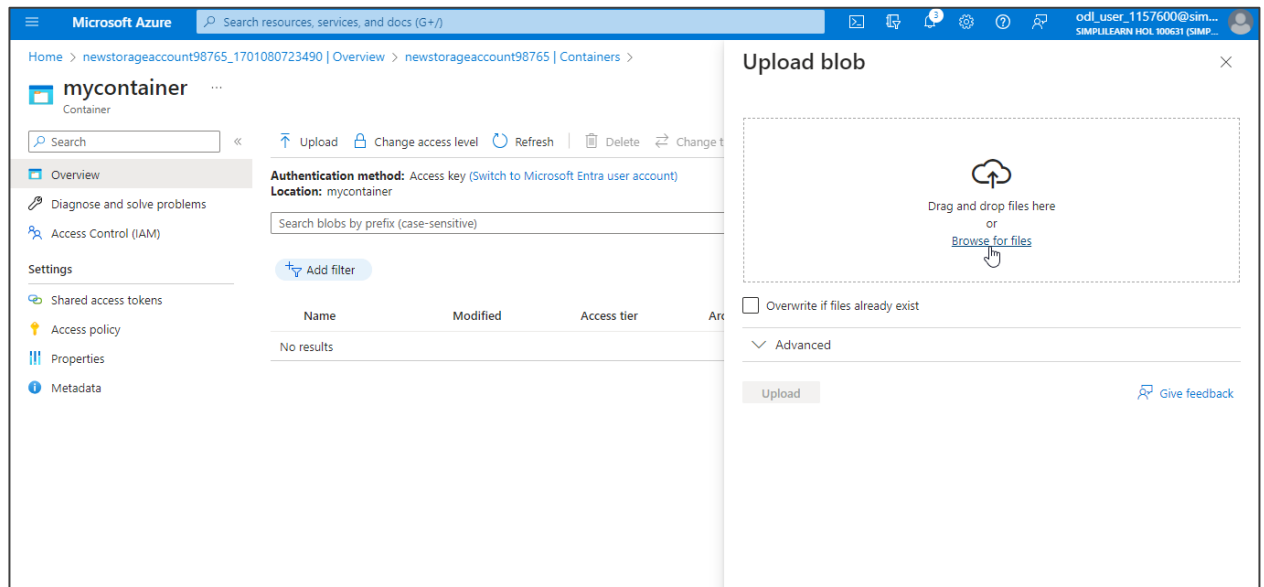
2.11 Click on the created container (mycontainer)



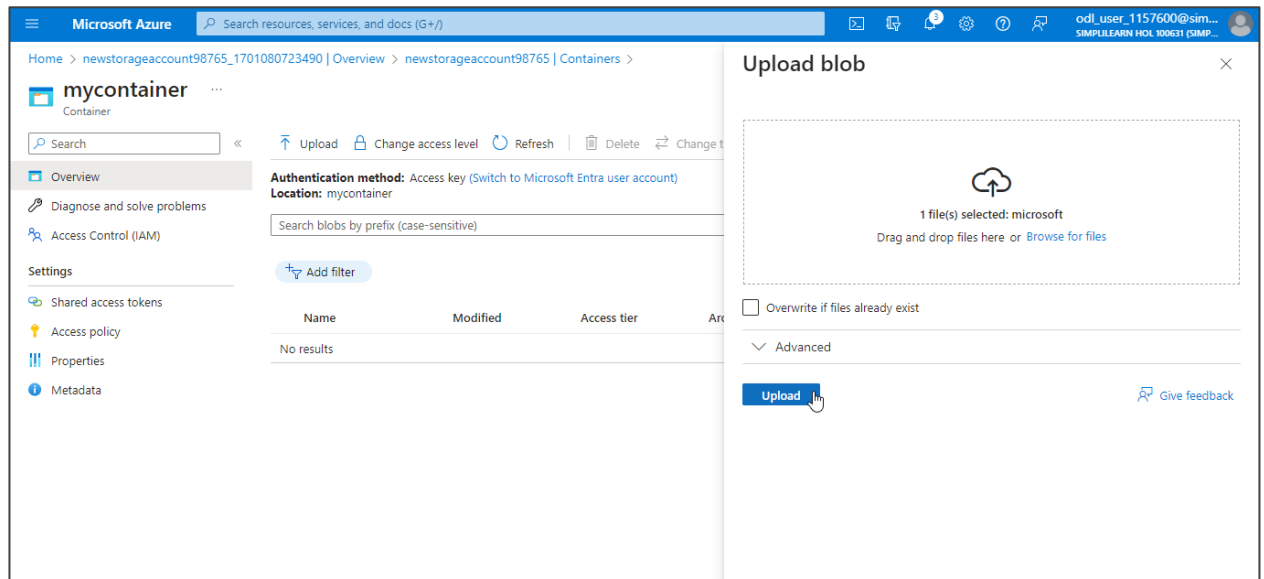
2.12 In the mycontainer page, click Upload



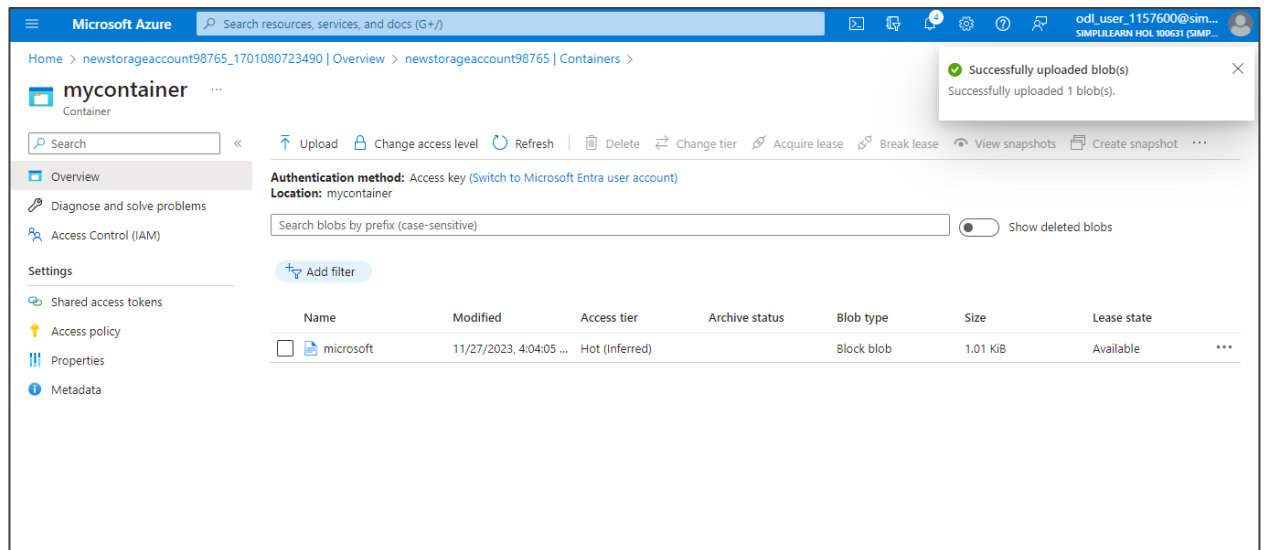
2.13 In the **Upload blob** page, click on **Browse for files** and select a file



2.14 Click **Upload**

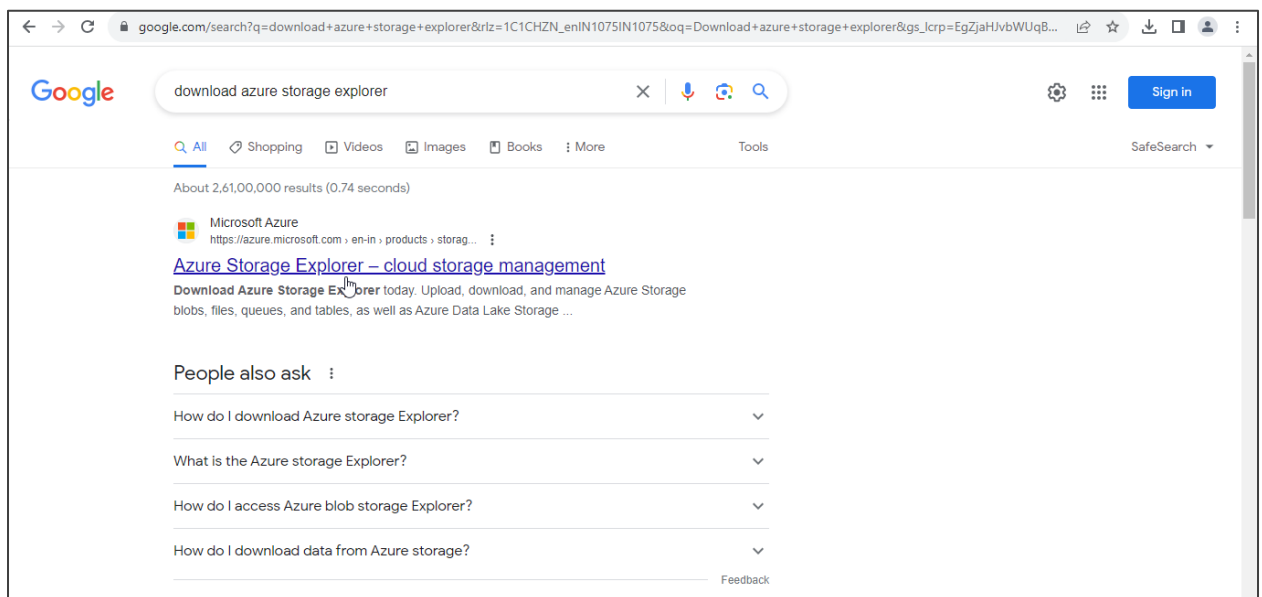


The blob has been uploaded as shown below:

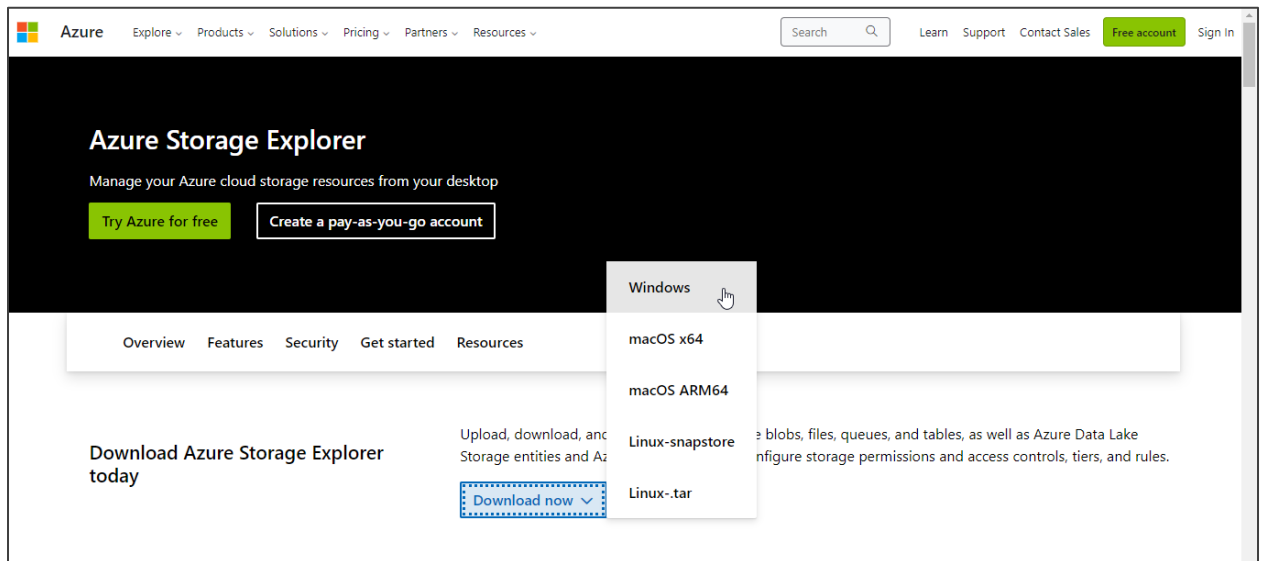


Step 3: Download Azure Storage Explorer

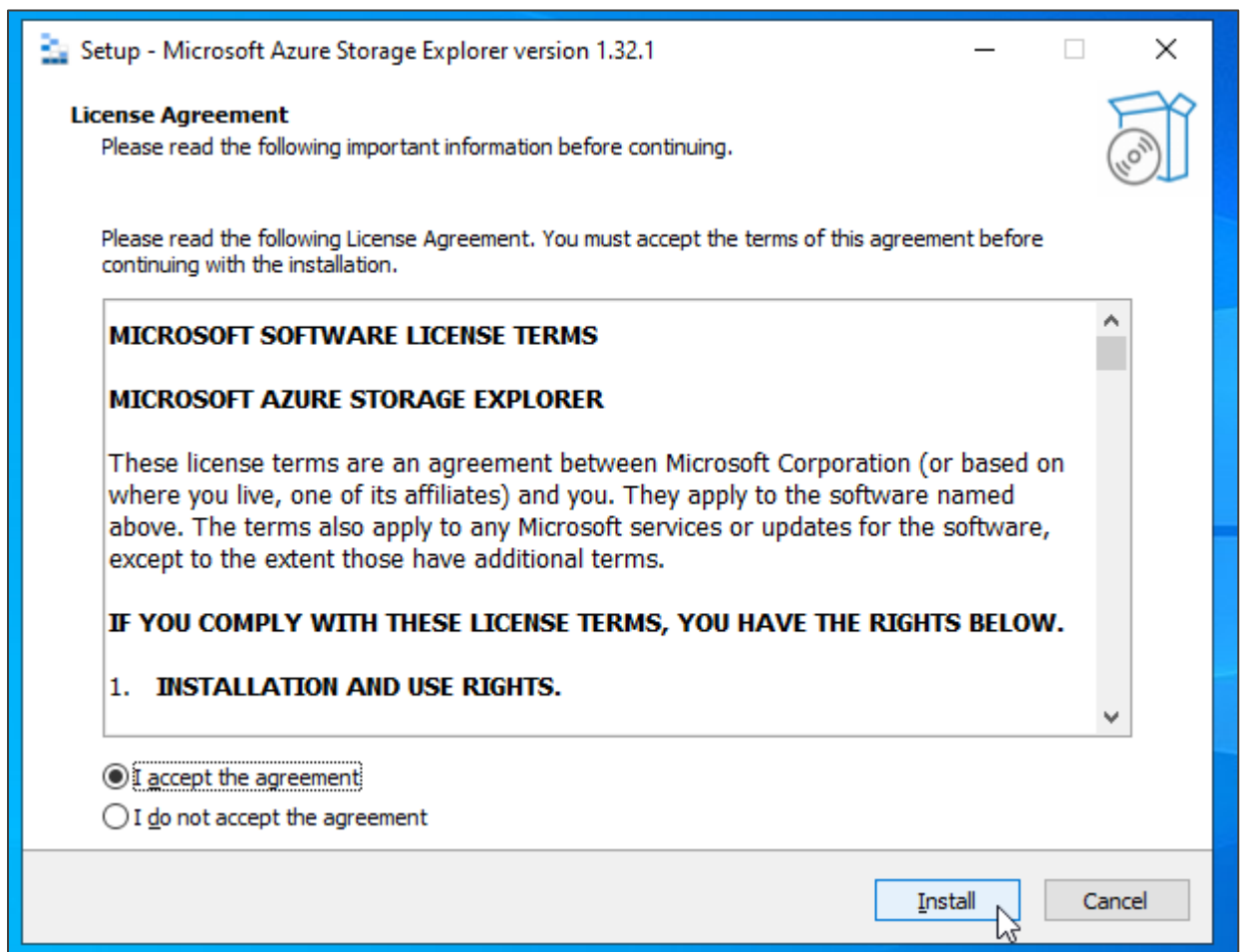
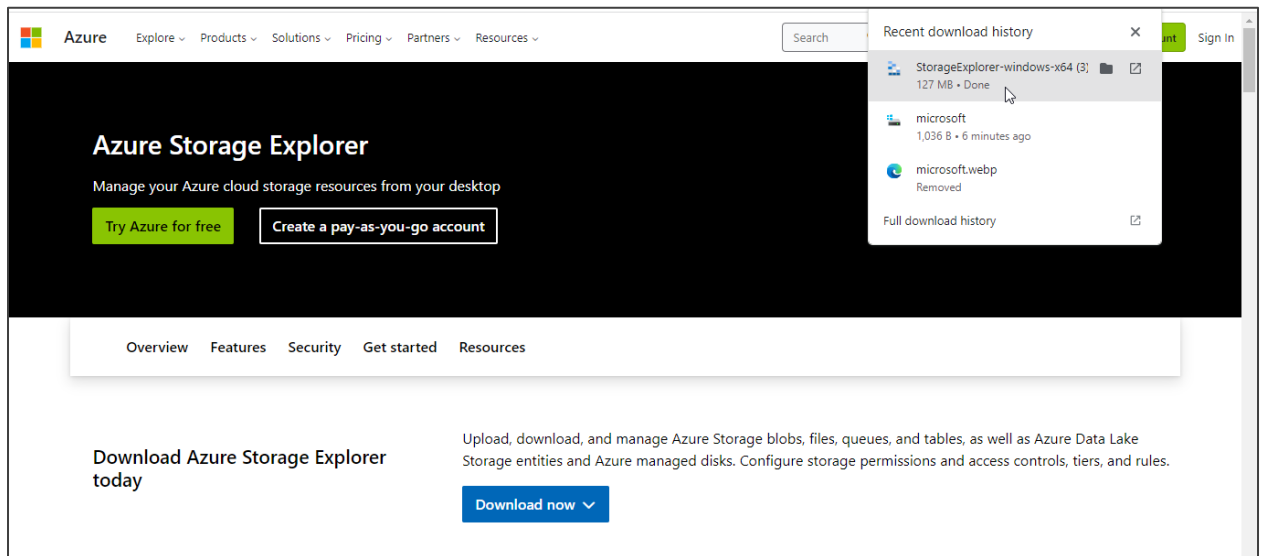
3.1 Open a new browser, search for **download azure storage explorer**, and select **Azure Storage Explorer – cloud storage management**



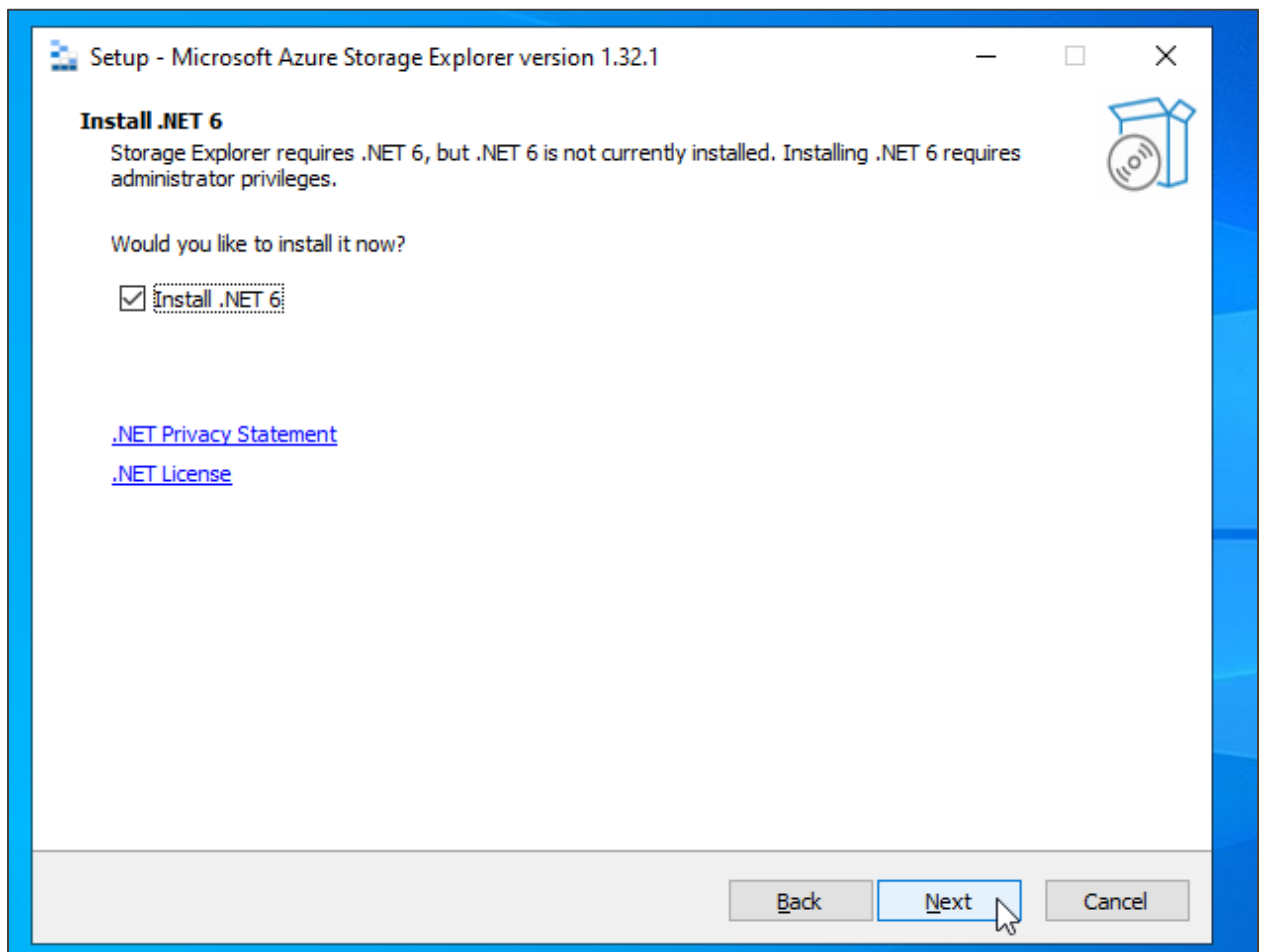
3.2 Click on **Download now** and select **Windows** from the drop-down list

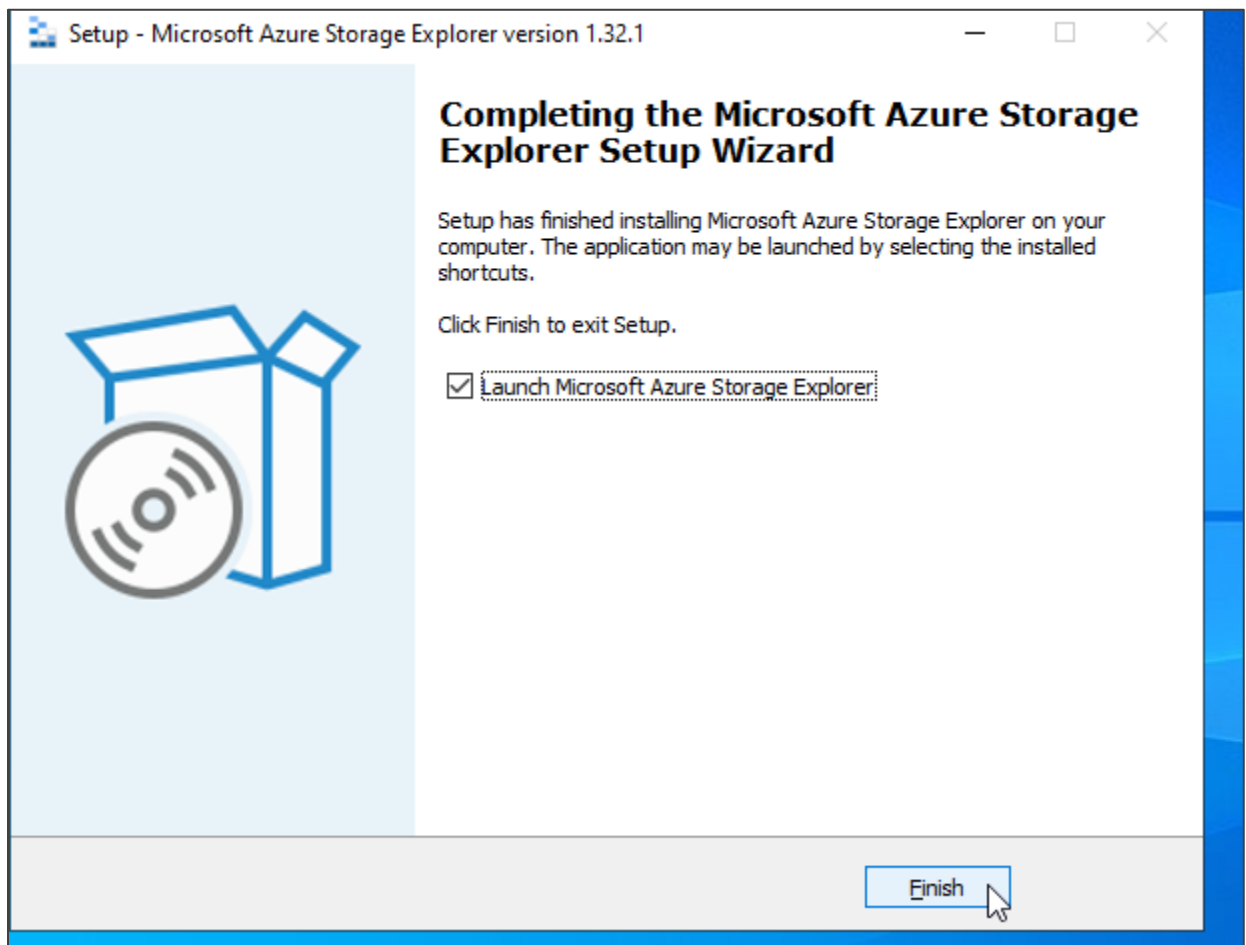


3.3 Select the downloaded file (StorageExplorer-windows-x64) and click Install

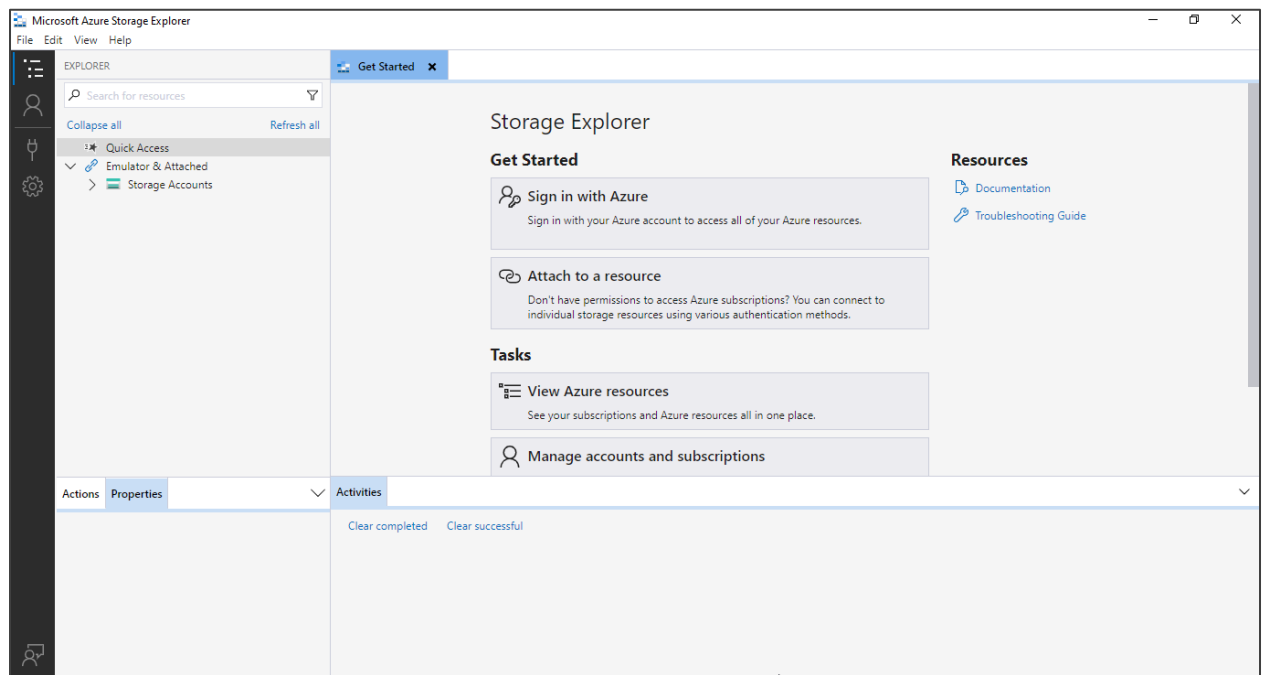


3.4 Check the **Install .NET 6** box and click **Next**

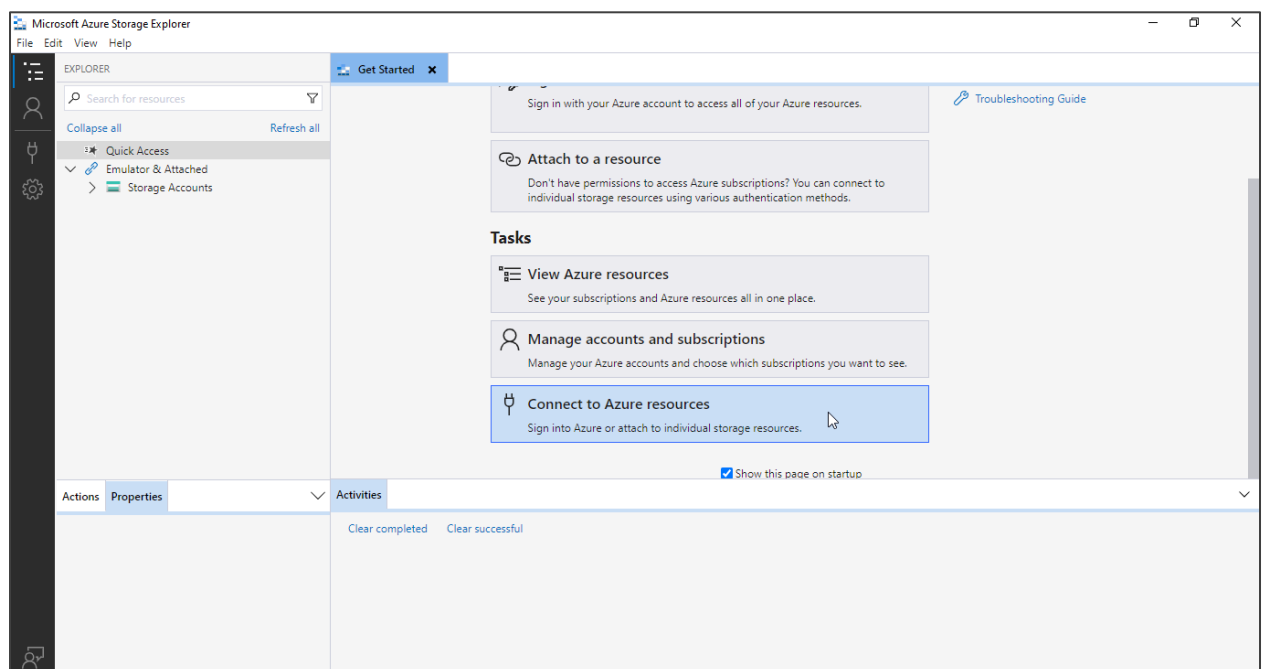


3.5 Click **Finish**

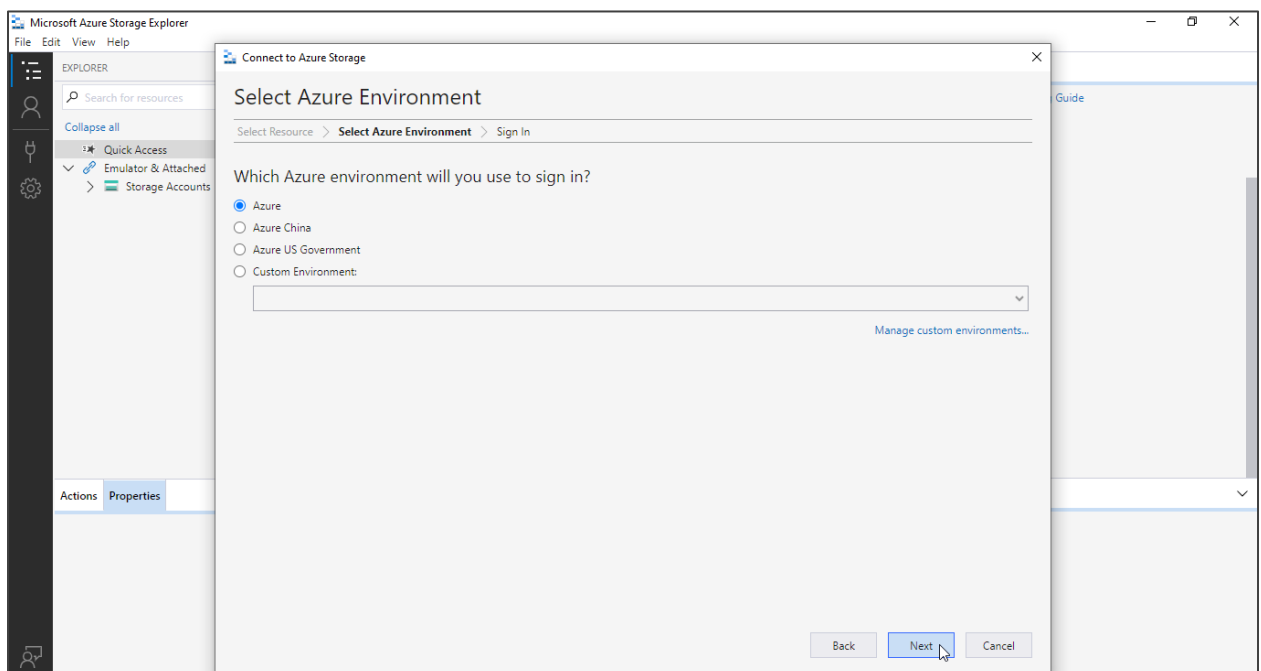
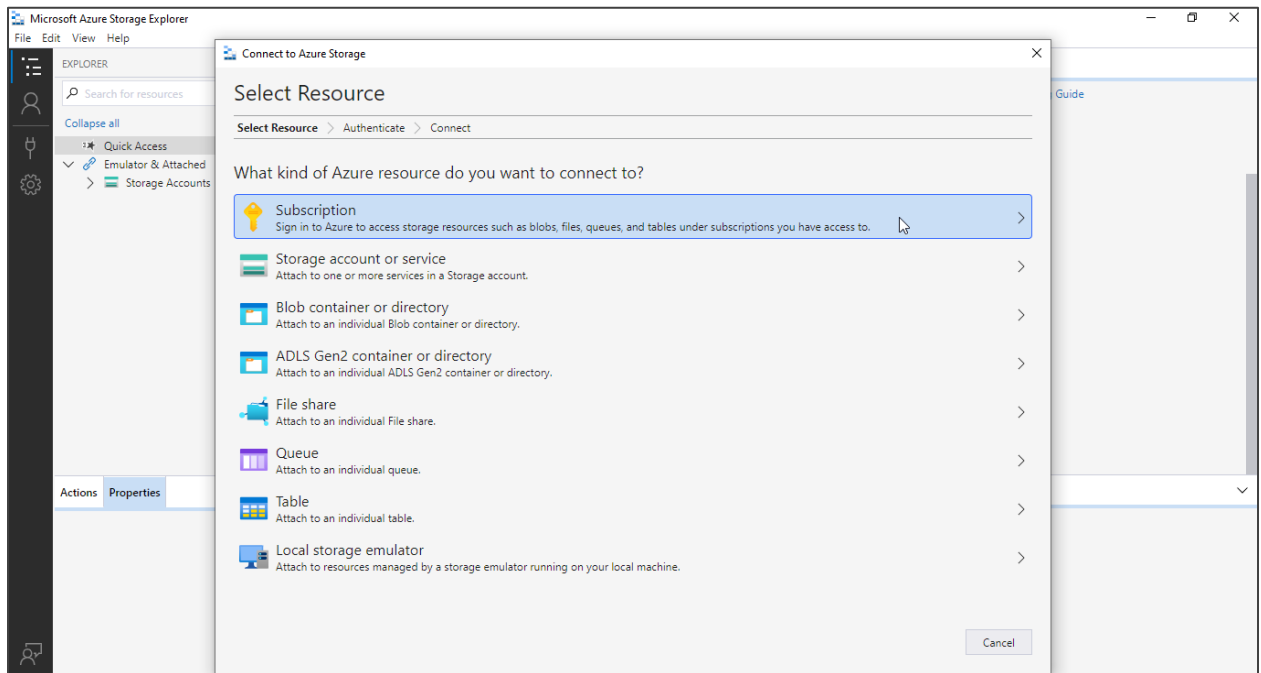
The **Microsoft Azure Storage Explorer** window appears as shown below:



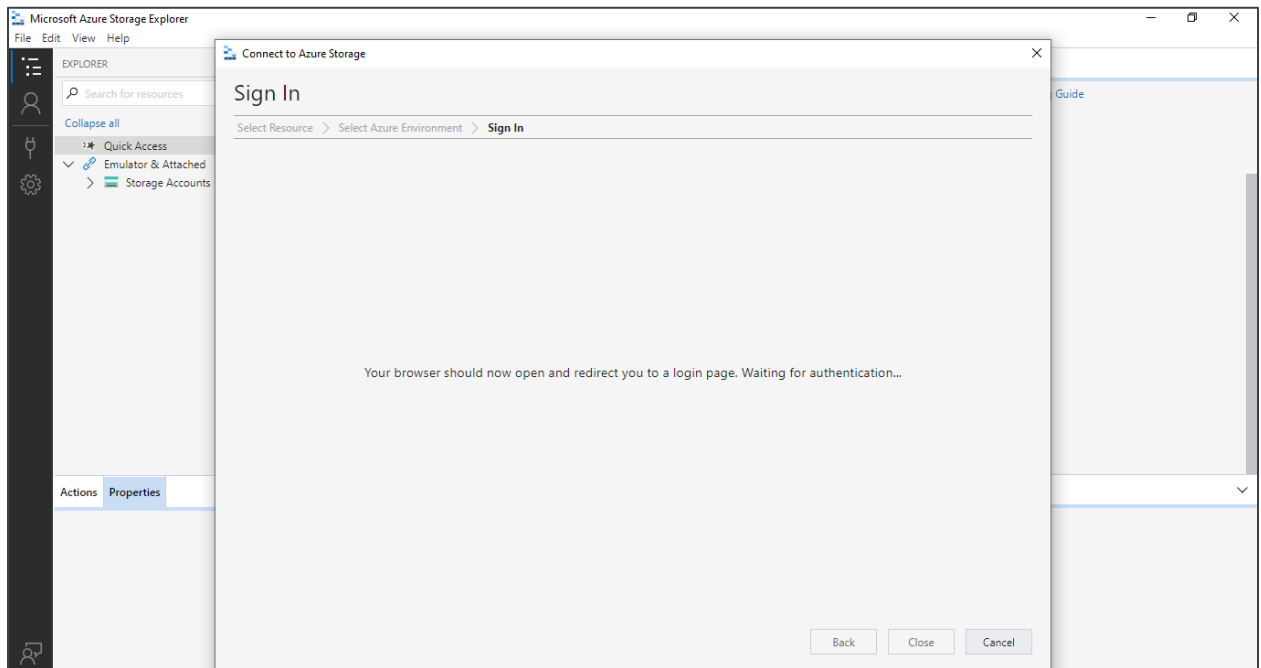
3.6 Click on **Connect to Azure resources**



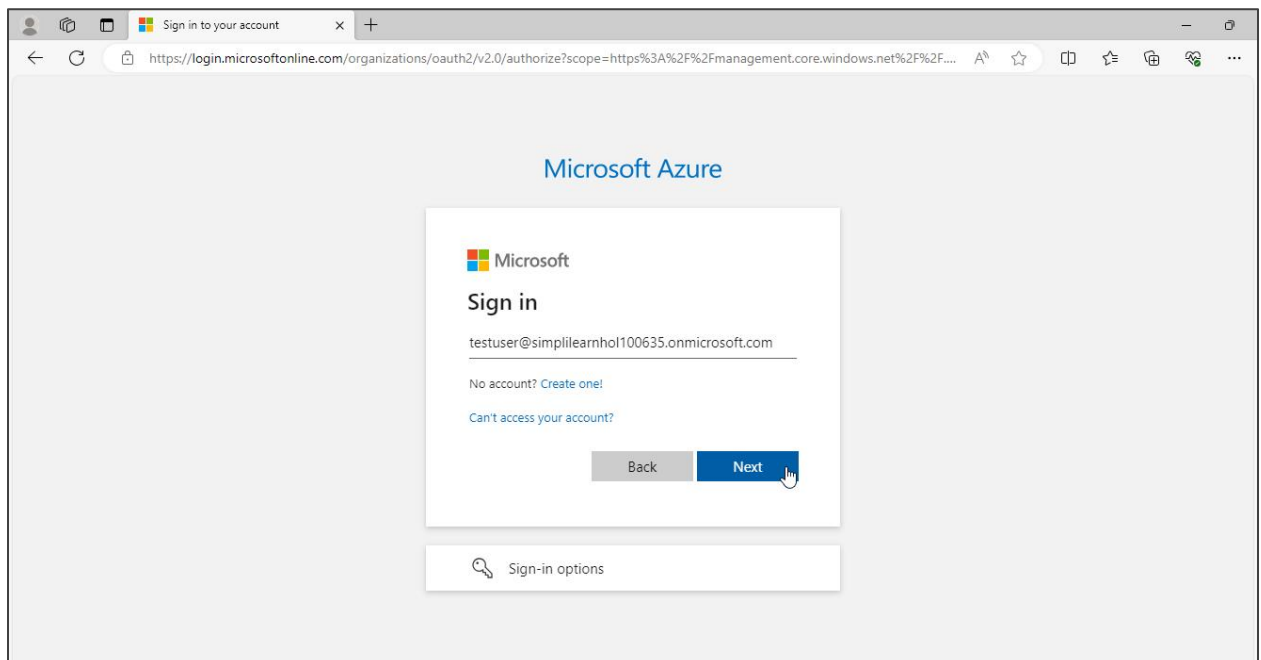
3.7 Select **Subscription** and click **Next**

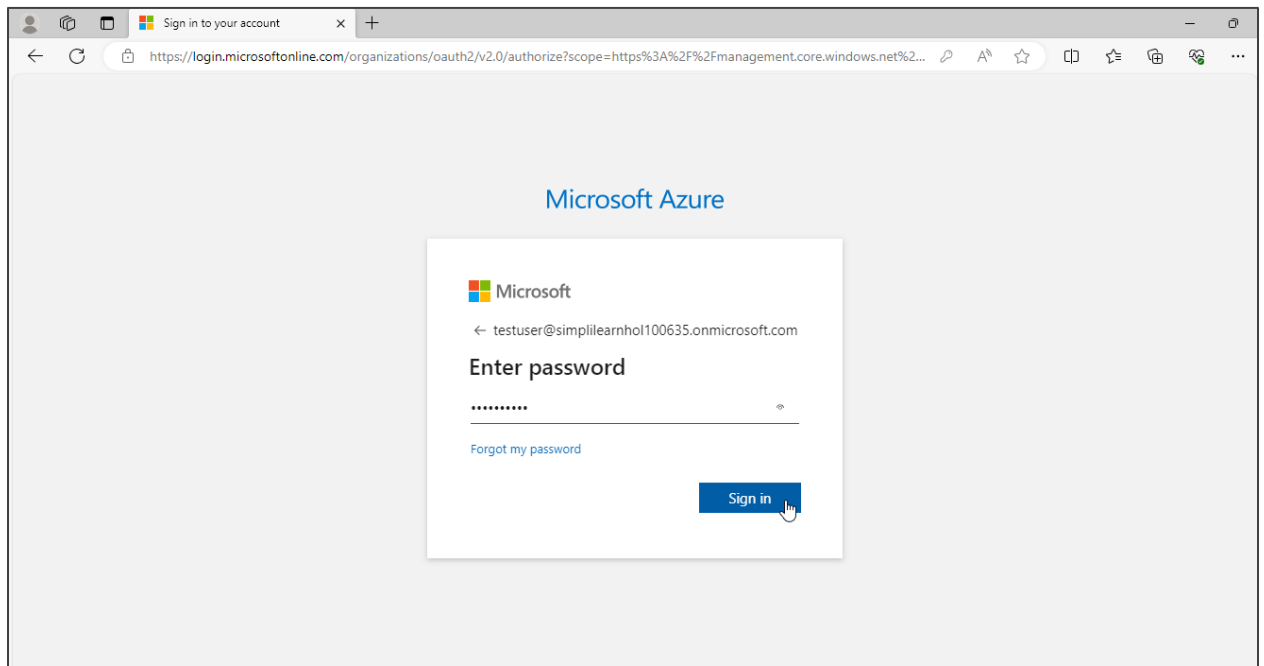


The following screen appears:

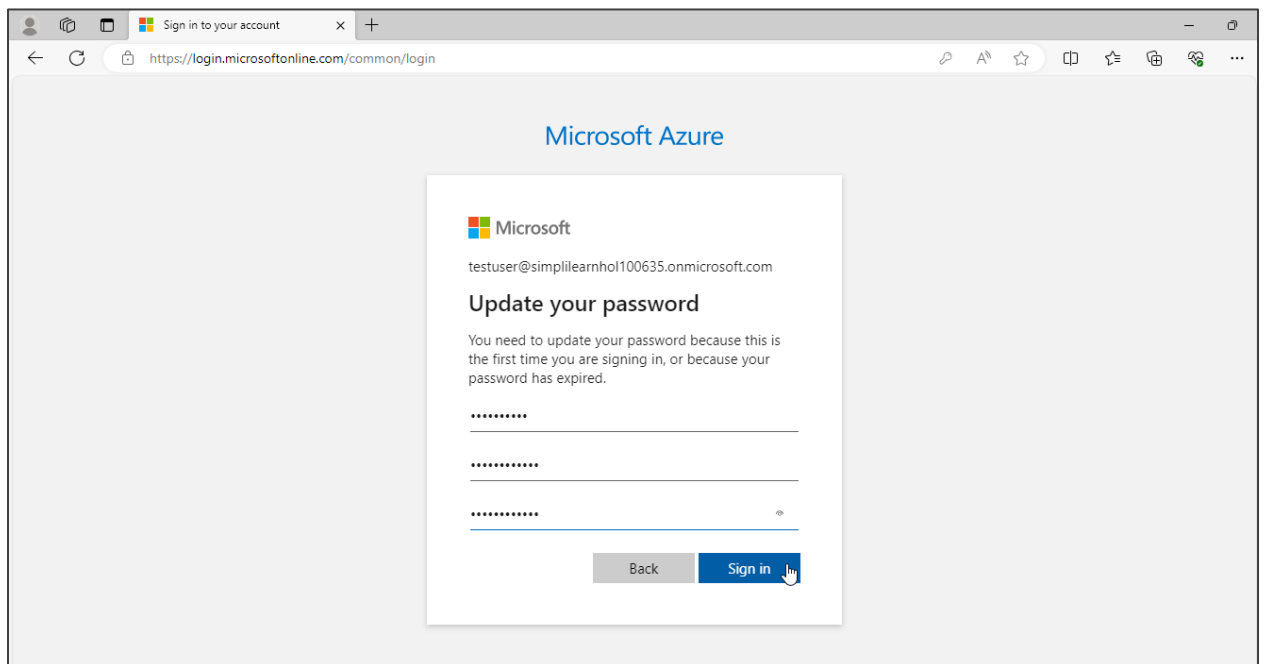


3.8 Log in using the credentials of the newly created user (**testuser01**) from Microsoft Entra ID

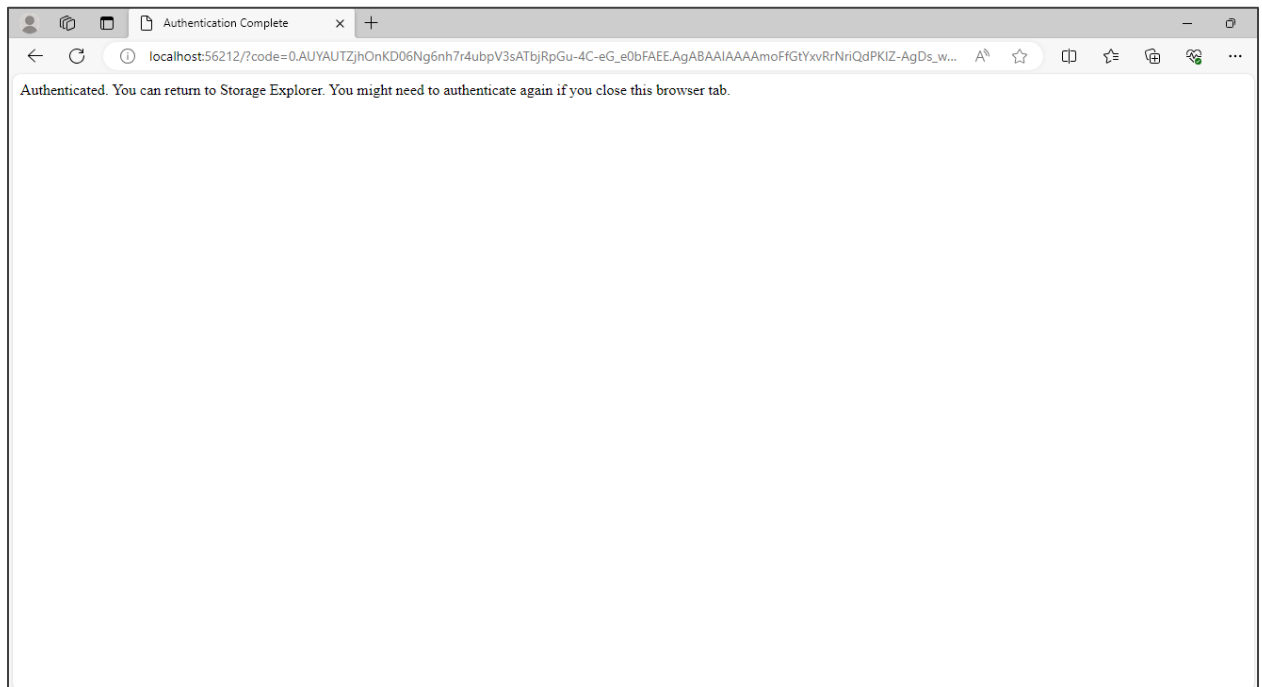




3.9 Update your password and click **Sign in**

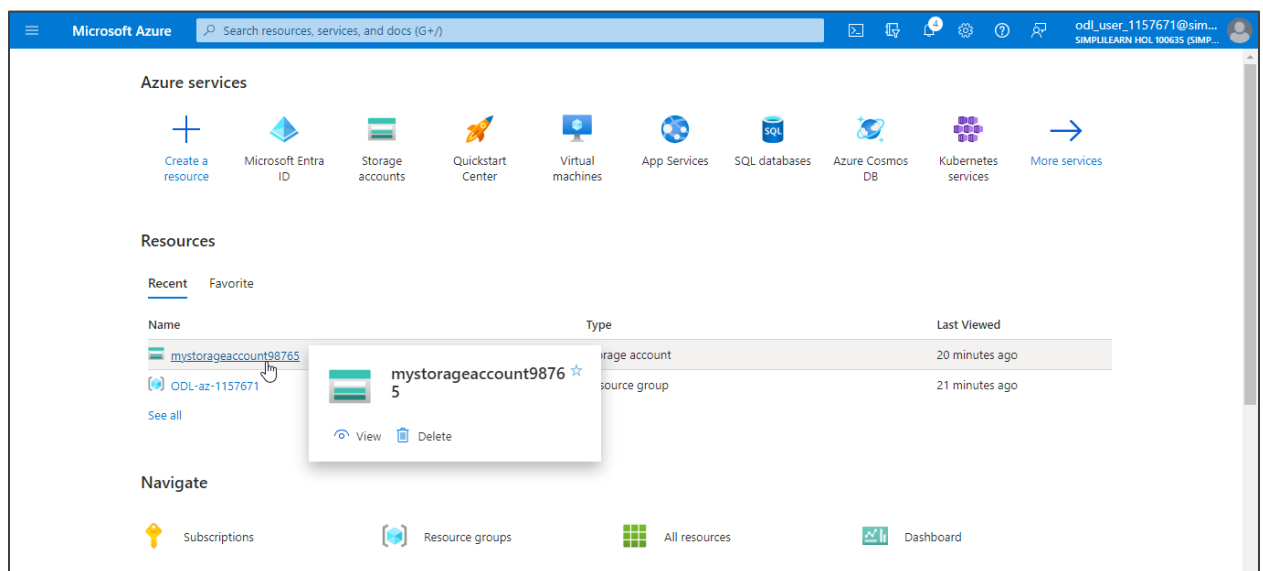


The following output appears:

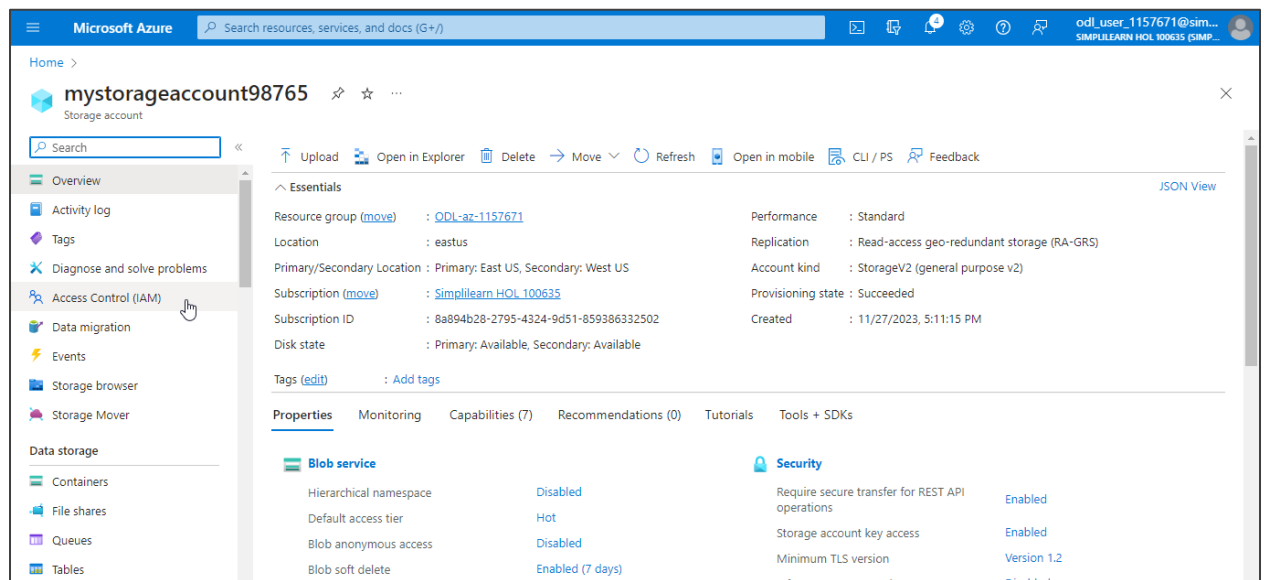


Step 4: Add Microsoft Entra ID authentication for storage

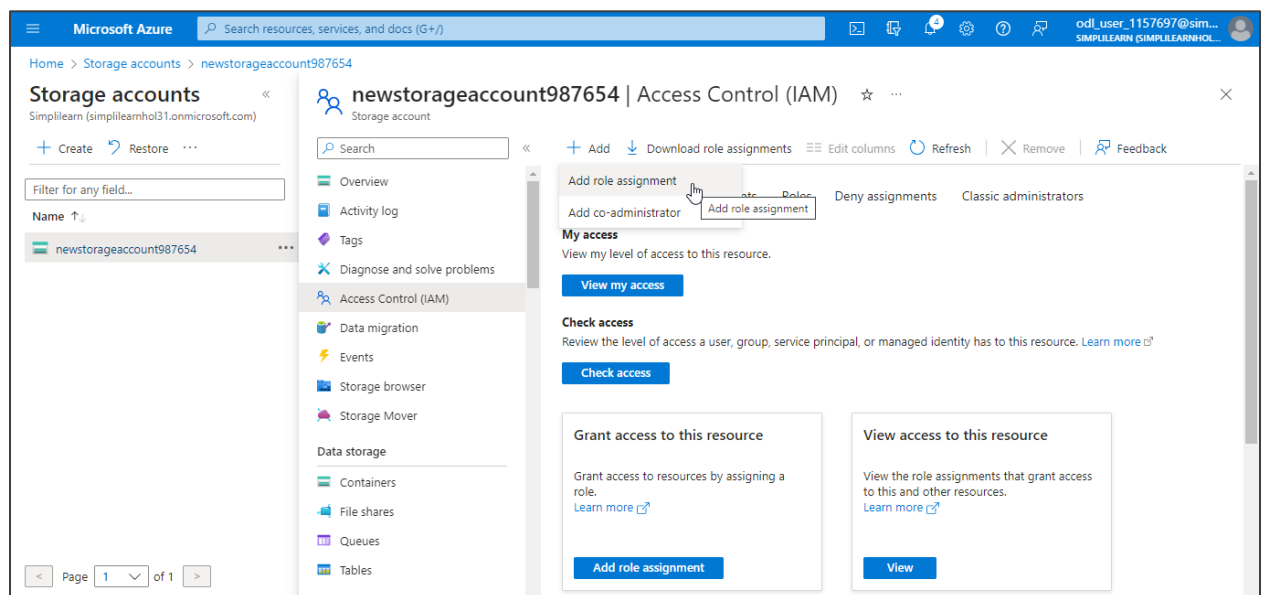
4.1 Navigate back to the **Microsoft Azure** home page and select the storage account that was created (**mystorageaccount98765**)



4.2 In the mystorageaccount98765 page, click **Access Control (IAM)**



4.3 In the **Access Control (IAM)** page, click **+ Add** and then select **Add role assignment**



4.4 In the **Add role assignment** page, navigate to the **Role** tab, select **Storage Blob Data Reader**, and click on **Next**

The screenshot shows the 'Add role assignment' page in the Microsoft Azure portal. The 'Role' tab is active, and the 'Storage Blob Data Reader' role is selected from the list. The 'Next' button is highlighted with a mouse cursor.

Name	Description	Type	Category	Details
Defender for Storage Data Scanner	Grants access to read blobs and update index tags. This role is used by the data scanner of Defender f...	BuiltinRole	None	View
Storage Blob Data Contributor	Allows for read, write and delete access to Azure Storage blob containers and data	BuiltinRole	Storage	View
Storage Blob Data Owner	Allows for full access to Azure Storage blob containers and data, including assigning POSIX access con...	BuiltinRole	Storage	View
Storage Blob Data Reader	Allows for read access to Azure Storage blob containers and data	BuiltinRole	Storage	View

Showing 1 - 4 of 4 results.

Buttons: Review + assign, Previous, **Next**, Next

4.5 Navigate to the **Members** tab and click **+ Select members**

The screenshot shows the 'Add role assignment' page in the Microsoft Azure portal, with the 'Members' tab active. The 'Storage Blob Data Reader' role is selected. The '+ Select members' button is highlighted with a mouse cursor.

Selected role: Storage Blob Data Reader

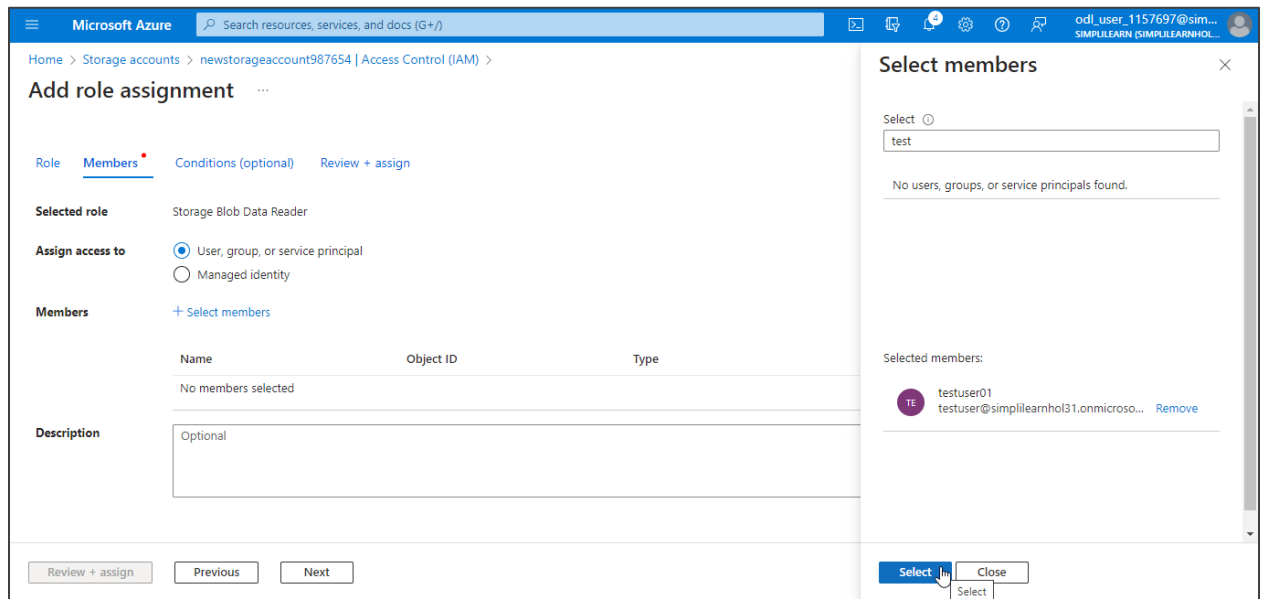
Assign access to: ☒ User, group, or service principal ☐ Managed identity

Members: [+ Select members](#)

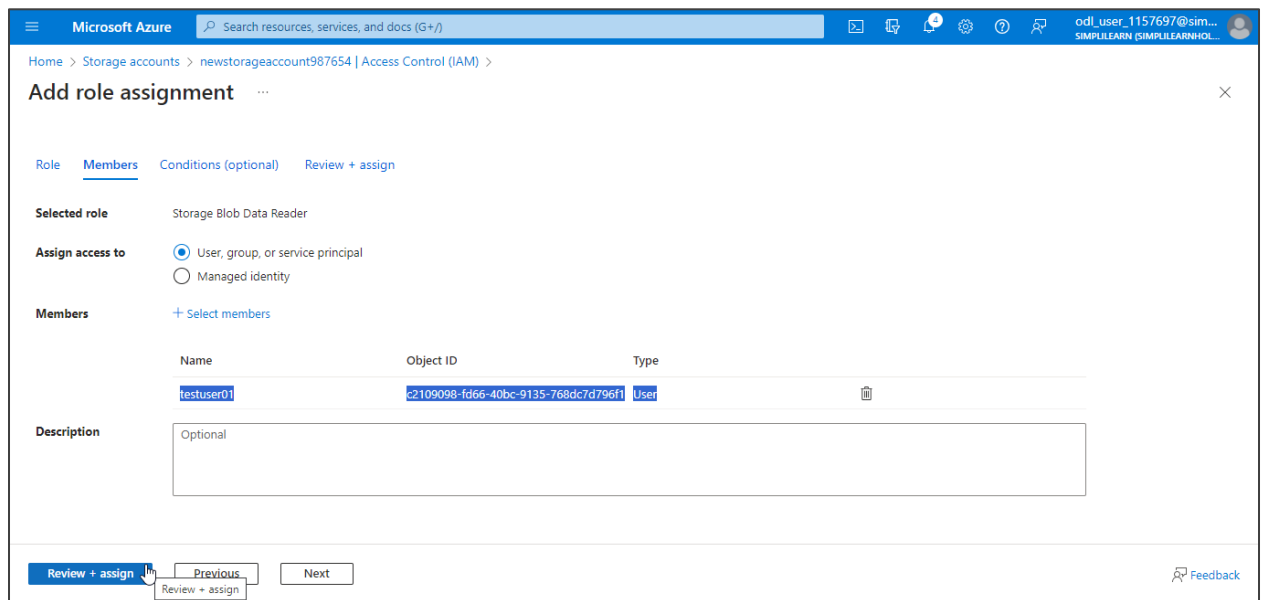
Name	Object ID	Type
No members selected		

Description: Optional

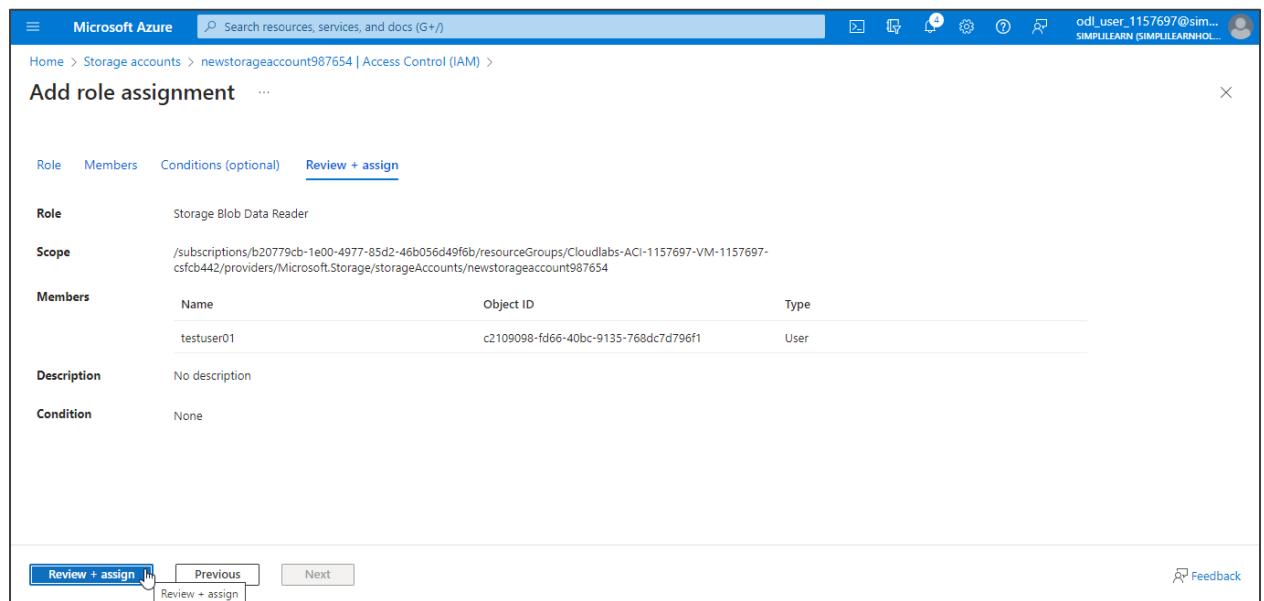
4.6 In the **Select members** page, search for and select **testuser01**, and click **Select**



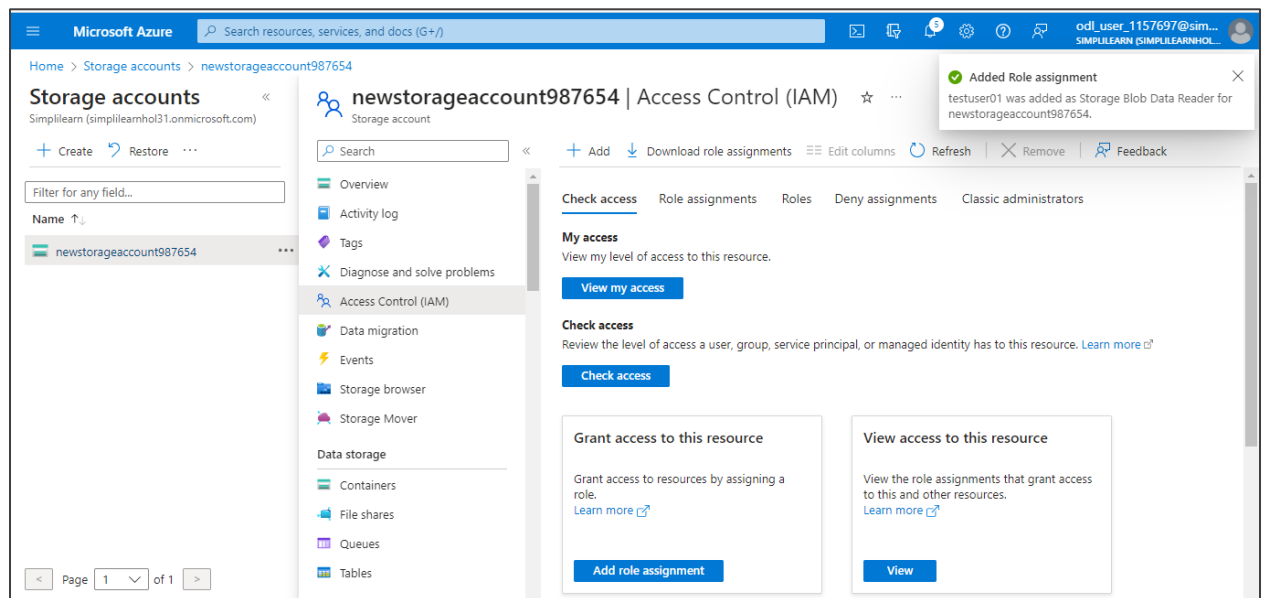
4.7 Click **Review + assign**



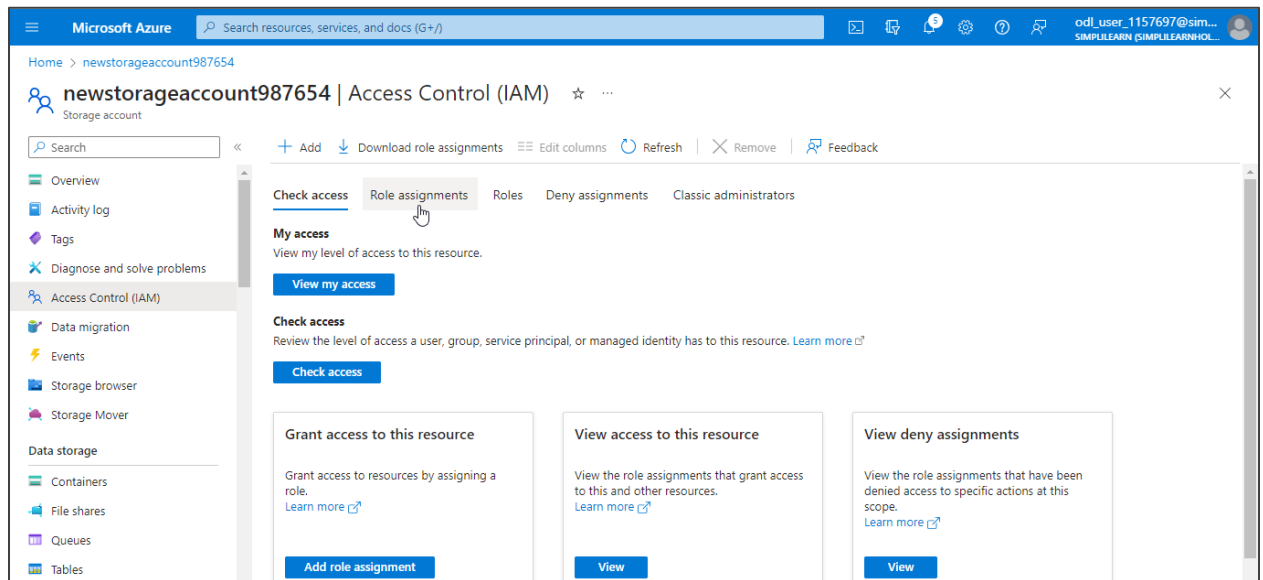
4.8 In the **Review + assign** tab, click **Review + assign**



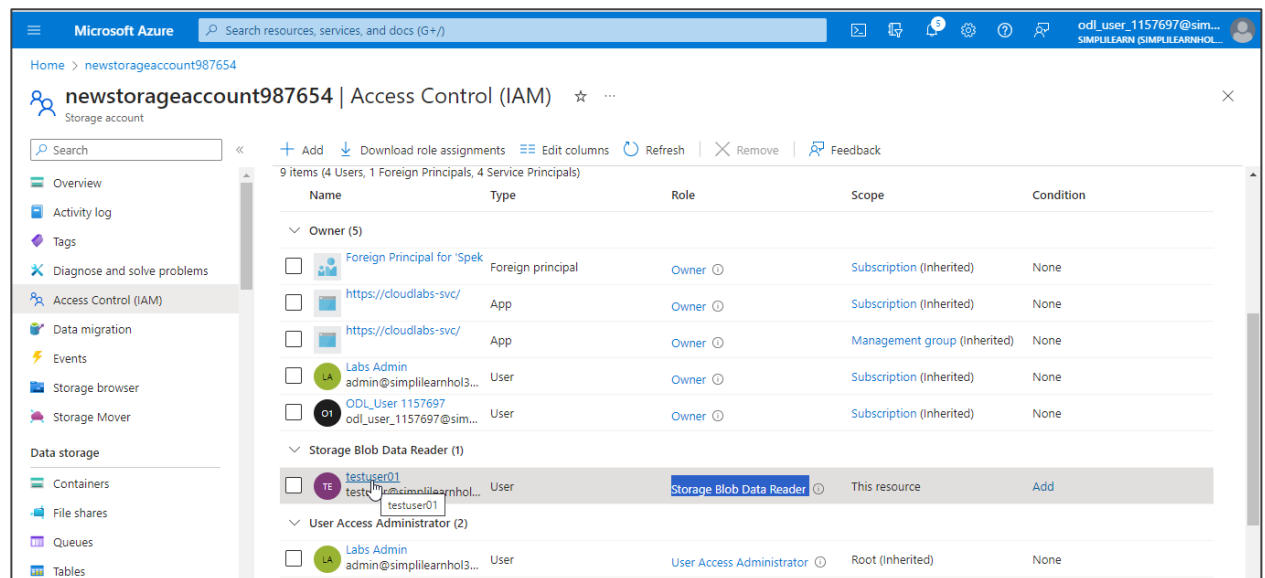
The role assignment has been added as shown below:



4.9 In the **Access Control (IAM)** page, click on **Role assignments** tab



4.10 Scroll down and click on **testuser01**



4.11 In the **testuser01** page, click on **Azure role assignments** under the **Manage** section

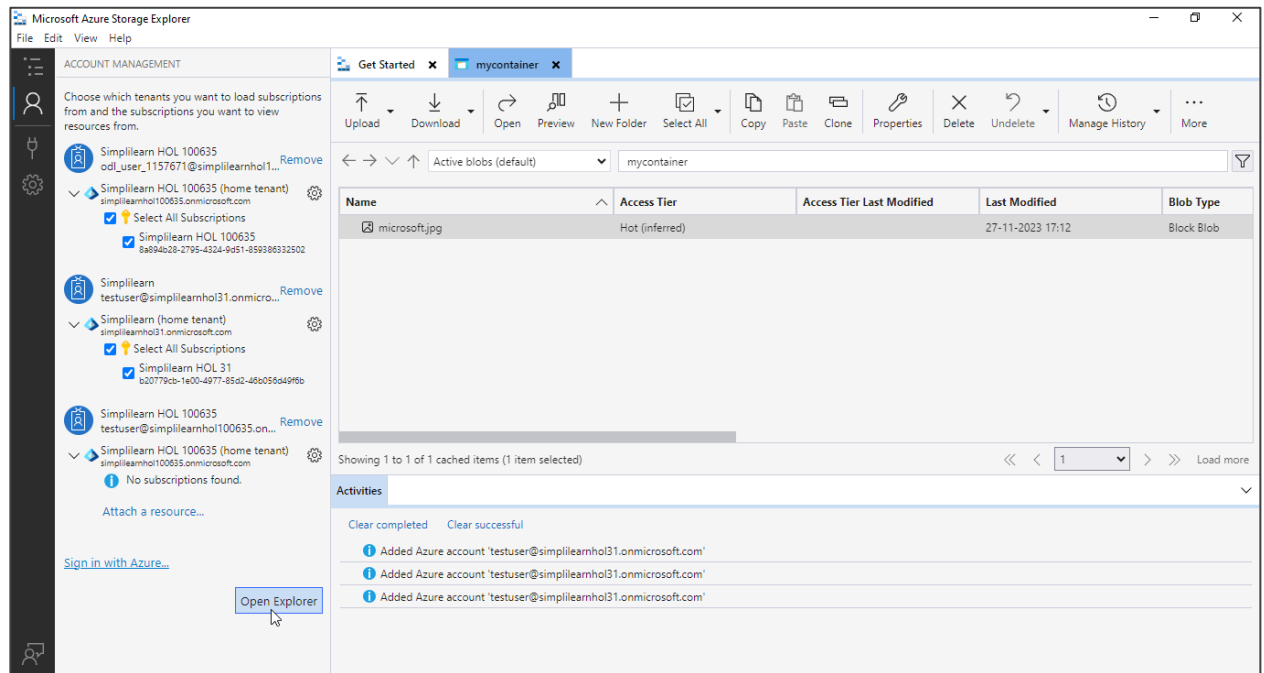
The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area displays the 'testuser01' user profile, including basic information like email, object ID, and creation date. The left-hand navigation pane is expanded, showing the 'Manage' section with various options. The 'Azure role assignments' option is highlighted with a mouse cursor.

The **Storage Blob Data Reader** role has been successfully assigned as shown below:

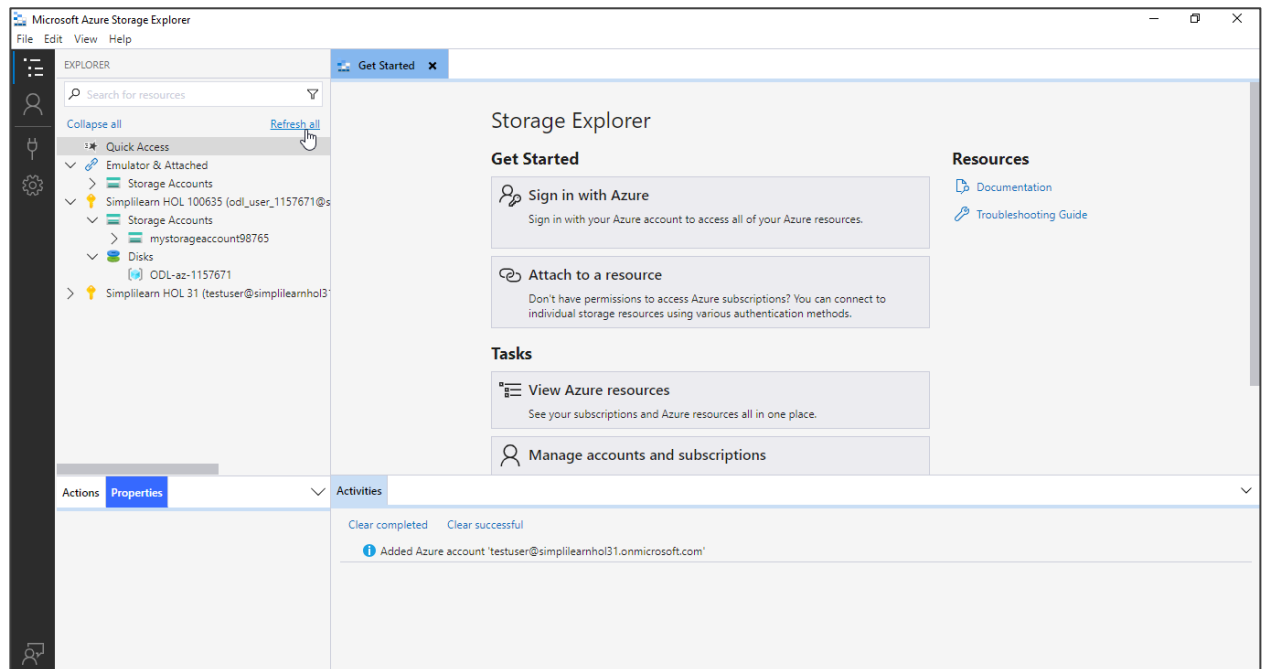
The screenshot shows the 'testuser01 | Azure role assignments' page in the Microsoft Azure portal. The page displays a table of assigned roles. The 'Storage Blob Data Reader' role is assigned to the 'newstorageaccount987654' storage account. A tooltip is visible over the role name.

Role	Resource Name	Resource Type	Assigned To	Condition
Storage Blob Data Reader	newstorageaccount987654	Storage account	testuser01	None

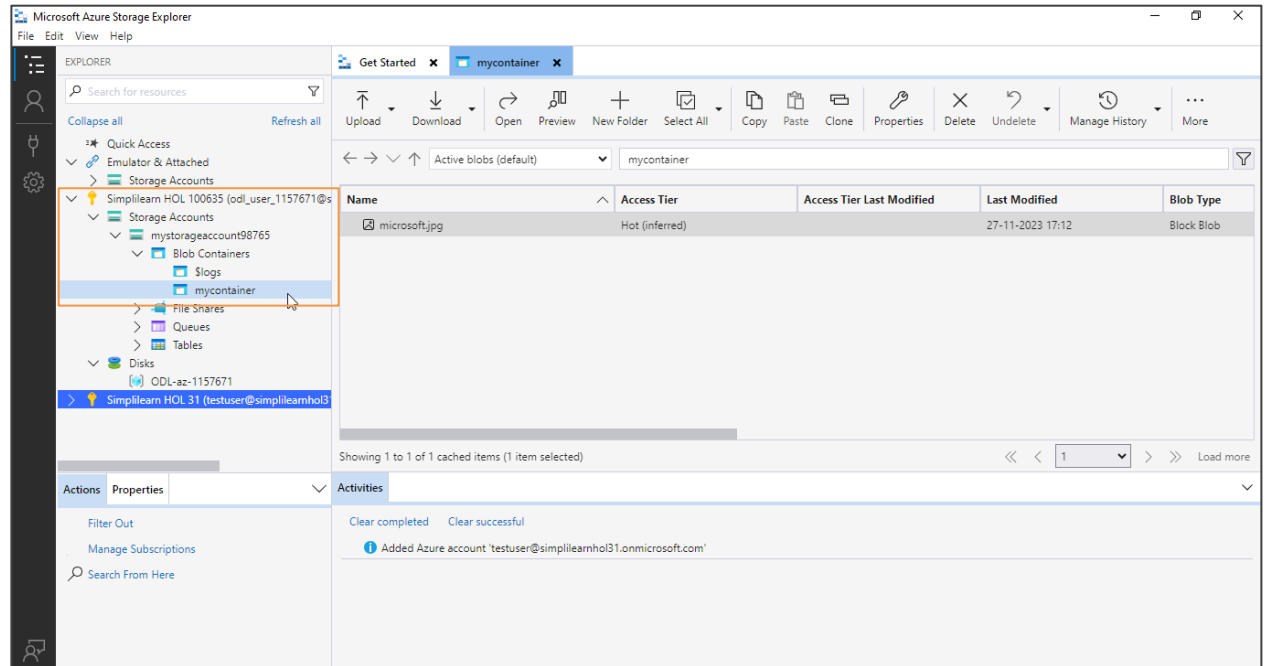
4.12 Navigate back to the Microsoft Azure Storage Explorer and click Open Explorer



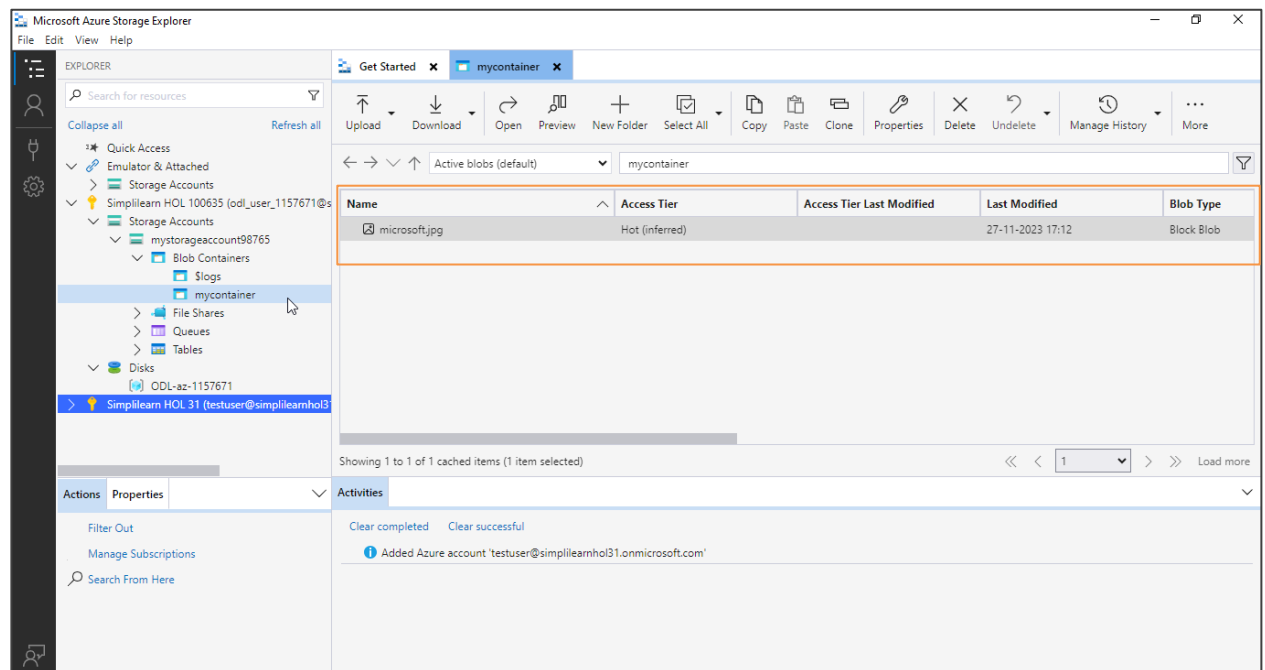
4.13 Click Refresh all



4.14 Click on the drop-down of **Simplilearn HOL 100635** and click on the container that was created (**mycontainer**)



The uploaded blob (**microsoft.jpg**) appears as shown below:



By following these steps, you have successfully created a guest user in Microsoft Entra ID and granted them relevant access, allowing access to Azure storage accounts using Storage Explorer.